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Chapter #7: Building Blocks of Integrated Circuit Amplifiers

from **Microelectronic Circuits** Text

by Sedra and Smith

Oxford Publishing

Introduction

- **IN THIS CHAPTER WE WILL LEARN**
  - The basic integrated-circuit (IC) design philosophy and how it differs from that for discrete-circuit design.
  - The basic gain cells of IC amplifiers, namely, the CS and CE amplifiers with current-source loads.
  - How to increase the gain realized in the basic gain cells by employing the principle of cascoding.
  - Analysis and design of the cascode amplifier and the cascode current source in both their MOS and bipolar forms.

Introduction

- IN THIS CHAPTER WE WILL LEARN
  - How current sources are used to bias IC amplifiers and how the reference current generated in one location is replicated at various other locations on the IC chip by using current mirrors.
  - Some ingenious analog circuit design techniques that result in current mirrors with vastly improved characteristics.
  - How to pair transistors to realize amplifiers with characteristics superior to those obtained from a single-transistor stage.

7.1. **Integrated Circuit Design Philosophy**

- Integrated-circuit fabrication technology imposes constraints on – and provides opportunities to – the circuit designer.
  - large capacitors are not available
  - very small capacitors are easy to fabricate
- One objective is to realize as many functions as possible using MOS transistors only.
- Reduction of device size is of great concern.
- In this text, focus is placed on CMOS circuit fabrication.

7.2. **The Basic Gain Cell**

- Two types of basic gain cells exist:
  - Common-source (CS)
  - Common-emitter (CE)
- Both are loaded with constant-current source.
  - This is done because of difficulties associated with fabrication of exact resistances.
  - It also facilitates increased gain.
- These circuits are referred to as current-source loaded / active loaded.

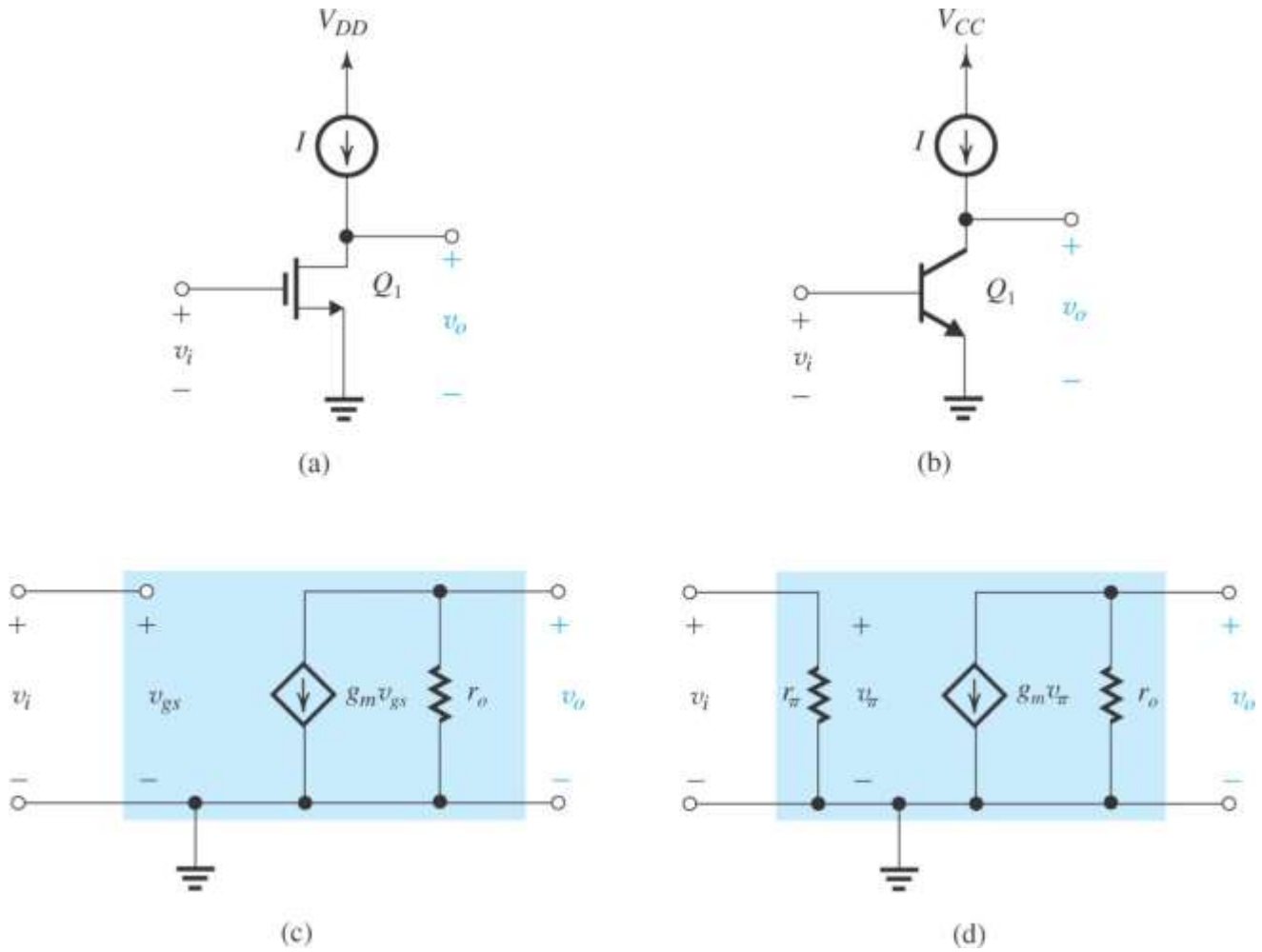


Figure 7.1 The basic gain cells of IC amplifiers: **(a)** current-source- or active-loaded common-source amplifier; **(b)** current-source- or active-loaded common-emitter amplifier; **(c)** small-signal equivalent circuit of **(a)**; and **(d)** small-signal equivalent circuit of **(b)**.

7.2. **The Basic Gain Cell**

The following equations applied to active-loaded CS amplifier:

(7.1) $R_{in} = \infty$

(7.2) $A_{vo} = -g_m r_o$

(7.3) $R_o = r_o$

The following equations applied to active-loaded CE amplifier:

(7.4) $R_{in} = r_\pi$

(7.5) $A_{vo} = -g_m r_o$

(7.6) $R_o = r_o$

7.2.2. **The Intrinsic Gain**

- For the BJT, one can derive a formulate for the intrinsic gain $A_{vo} = g_m r_o$ using the formulas below.

$$(7.7)\ g_m = \frac{I_C}{V_T}$$

$$(7.8)\ r_o = \frac{V_A}{I_C}$$

$$(7.9)\ \text{intrinsic gain is: } A_0 = g_m r_o = \frac{V_A}{V_T}$$

7.2.2. **The Intrinsic Gain**

- A_0 is simply ratio of the Early Voltage (V_A) and thermal voltage (V_T)
  - The value of V_A ranges from 5V to 35V for modern technologies.
  - The value of A_0 ranges from 200V/V to 5000V/V, as such.
- There are three possible expressions for g_m, two are particularly useful here.

$$(7.10)\ g_m = \frac{I_D}{V_{OV}/2}$$

$$(7.11)\ g_m = \sqrt{2\mu_n C_{ox}\frac{W}{L}}\sqrt{I_D}$$

7.2.2. **The Intrinsic Gain**

The following equations apply to MOSFET:

$$(7.12)\ r_o = \frac{V_A}{I_D} = \frac{V_A' L}{I_D}$$

$$(7.13)\ A_0 = \frac{V_A}{V_{OV}/2}$$

$$(7.14)\ A_0 = \frac{2V_A' L}{V_{OV}}$$

$$(7.14)\ A_0 = \frac{V_A' \sqrt{2(\mu_n C_{ox})WL}}{\sqrt{I_D}}$$

7.2.2. **The Intrinsic Gain**

- The expression (7.13) is one most comparable to the BJT (7.9).
- Note the following points:
  - The quantity of $V_{OV}/2$ is a design parameter.
    - Its value has been decreasing with technological developments.
  - The numerator quantity is both process dependent (through V_A') and device dependent (through L).
    - Its value has been decreasing with technological developments.
  - From (7.14) we see that A_0 may be increased by using a longer MOSFET and operating at lower V_{OV}.
    - This is not without trade-offs (refer to textbook).

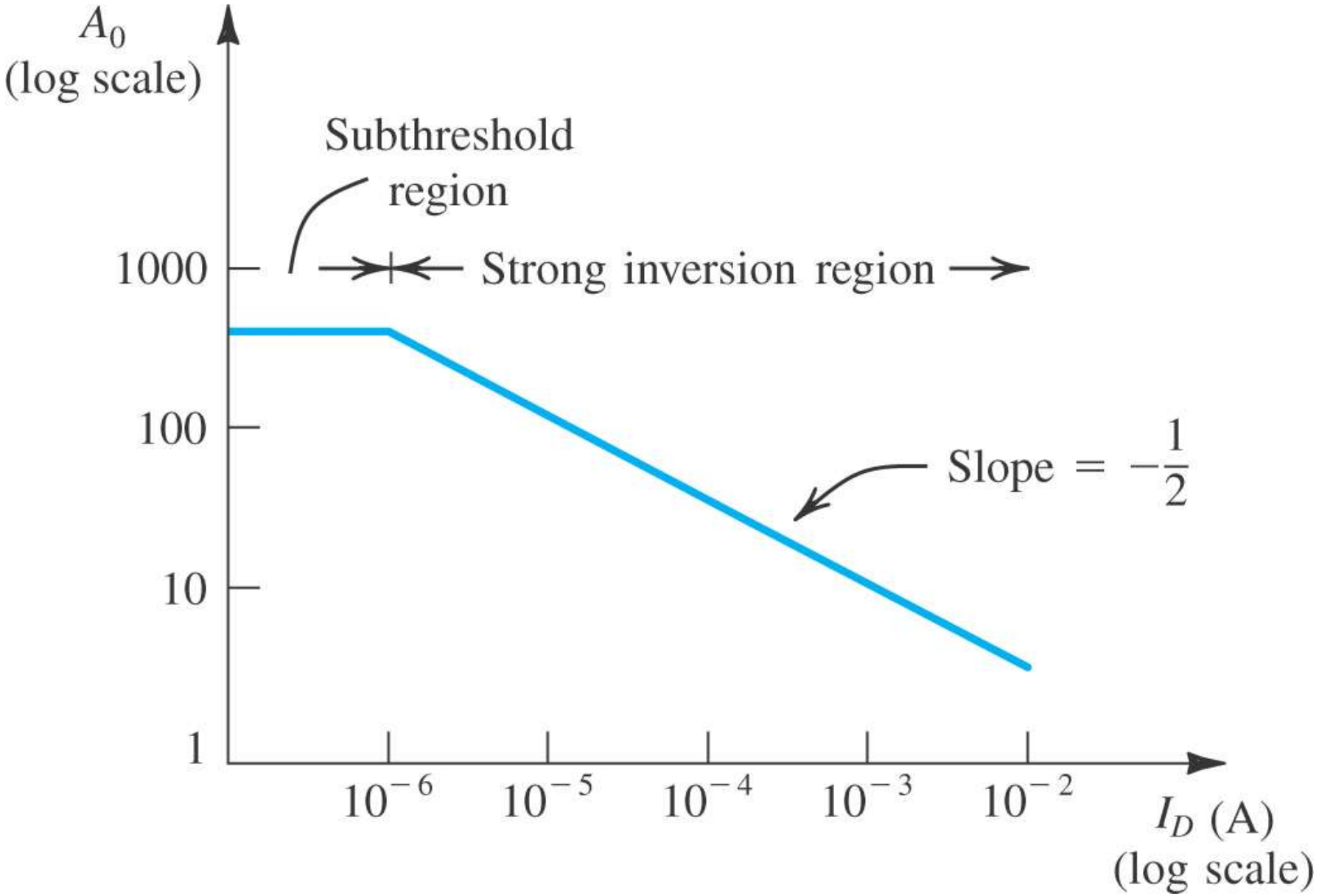


Figure 7.2 The intrinsic gain of the MOSFET versus bias current I_D. Outside the subthreshold region, this is a plot of $A_0 = V'A\ \sqrt{2\mu_n C_{ox} WL/I_D}$ for the case: $\mu_n C_{ox} = 20\ \mu A/V^2$, $V'A = 20\ V/\mu m$, $L = 2\ \mu m$, and $W = 20\ \mu m$.

7.2.3. **Effect of the Output Resistance of the Current-Source Load**

- The current-source load of the CS amplifier in Figure 7.1(a) can be implemented using a PMOS transistor biased in the saturation region to provide the required current *I*, as shown in Figure 7.3(a).

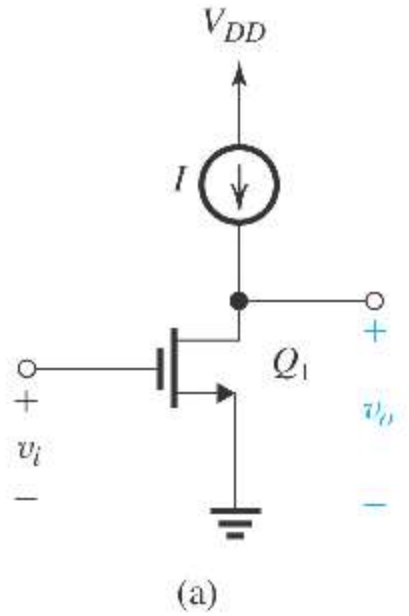


Figure 7.1(a)

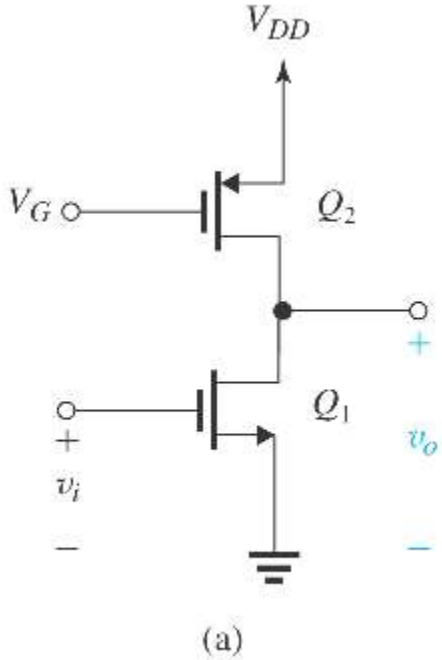


Figure 7.3(a)

7.2.3. Effect of the Output Resistance of the Current-Source Load

for PMOS implementation of active-loaded CS amplifier:

large-signal MOSFET model

$$(7.16)\quad I = \tfrac{1}{2}\left(\mu_p C_{ox}\right)\frac{W}{L}\left[V_{DD} - V_G - \left|V_{tp}\right|\right]^2$$

$$(7.17)\quad r_{o2} = \frac{\left|V_{A2}\right|}{I}$$

current-source no longer has infinite output resistance

$$(7.18)\quad A_v = \frac{v_o}{v_i} = -g_{m1}\left(r_{o1} \,||\, r_{o2}\right)$$

7.2.3. **Effect of the Output Resistance of the Current-Source Load**

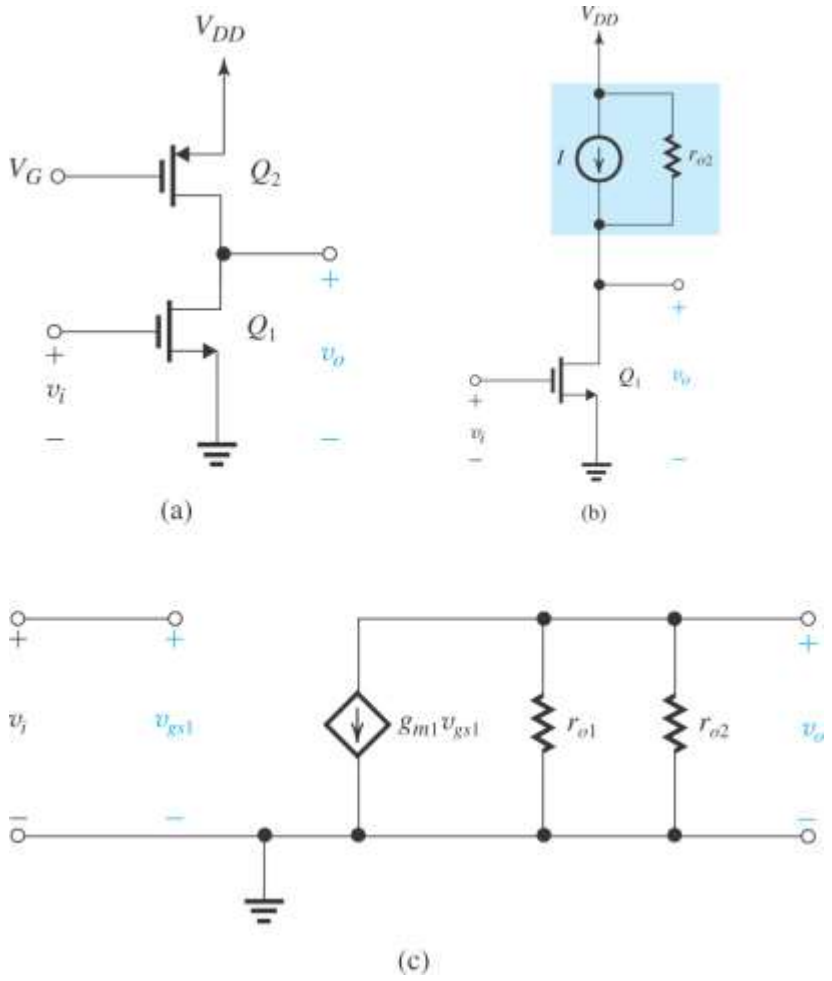
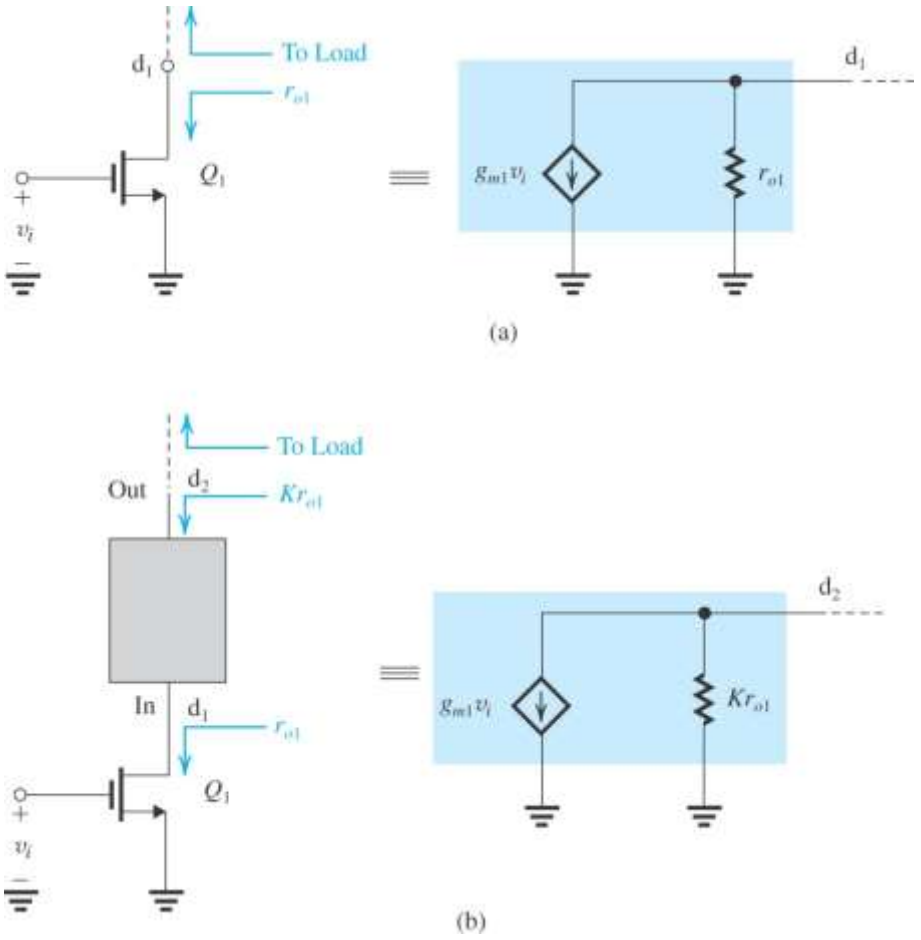


Figure 7.3 (a): The CS amplifier with the current-source load implemented with a *p*-channel MOSFET Q_2; **(b)** the circuit with Q_2 replaced with its large-signal model; and **(c)** small-signal equivalent circuit of the amplifier.

7.2.4. **Increasing the Gain of the Basic Cell**

Figure 7.5: To increase the voltage gain realized in the basic gain cell shown in **(a)**, a functional block, shown as a black box in **(b)**, is connected between d_1 and the load.

- **Q:** How can we increase the voltage gain obtained from basic gain cell?
  - **A:** Find a way to raise the level of output resistance.
  - **A:** Seek a circuit that passes the current $g_m v_i$ provided by the amplifying transistor right through.
    - But increases the resistance from r_o to a much larger value.



7.2.4. **Increasing the Gain of the Basic Cell**

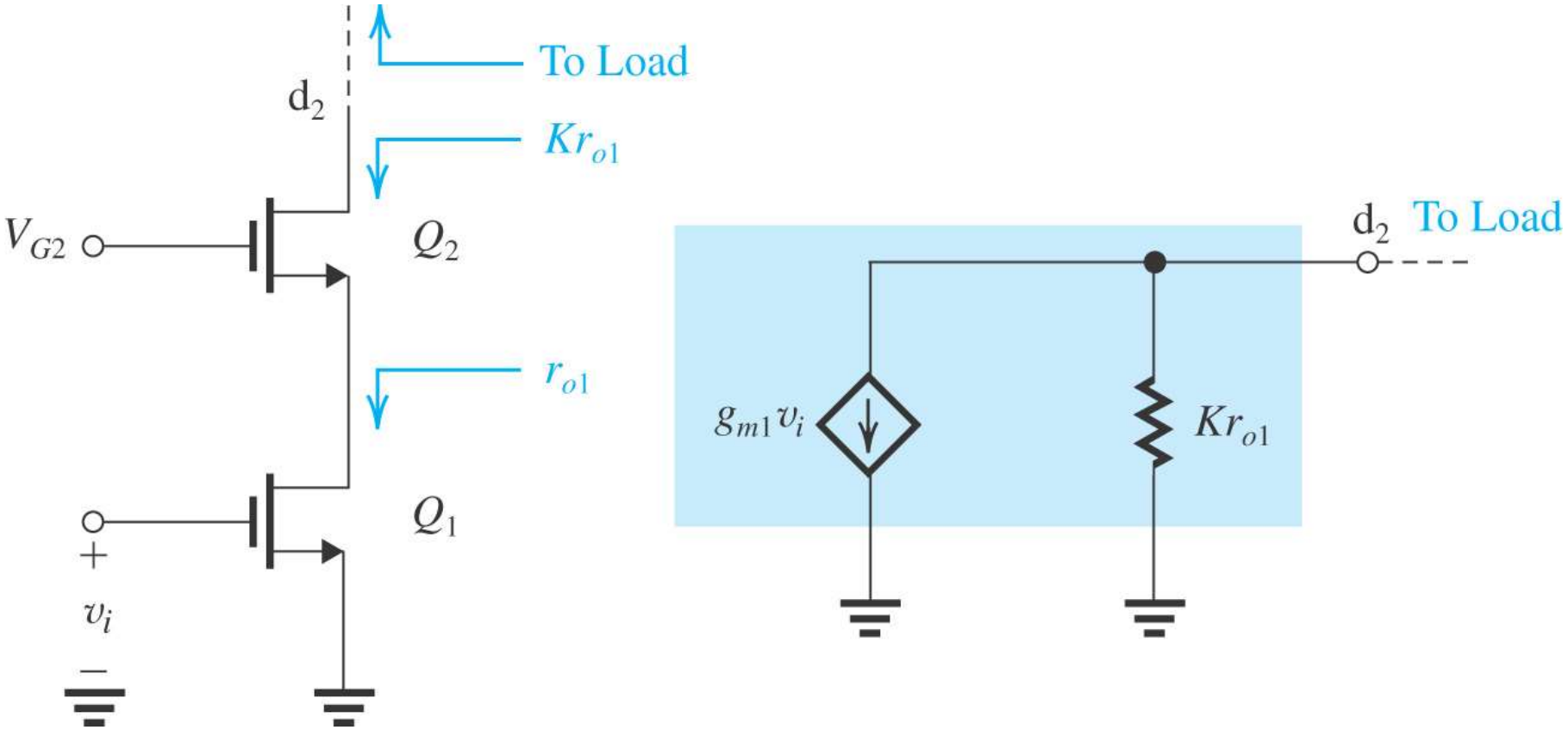
- The black box of Figure 7.5. is a **current buffer** – aka. a device which passes voltage but lowers resistance level.
- Two important observations should be made:
  - It is not sufficient to raise output resistance of amplifying transistor only.
  - Placing CG (or a CB) circuit on top of the CS (or CE) amplifying transistor to implement the current-buffering action is called **cascading**.

7.3. **The Cascode Amplifier**

- **Cascoding** refers to the use of the transistor connected in the common-gate (or common-base) configuration.
- Provides current buffering for the output of a common-source (or common-emitter) amplifying transistor.
- Figure 7.6. illustrates this technique for MOS case.

7.3. **The Cascode Amplifier**

Figure 7.6: The current-buffering action of Figure 7.5(a) is implemented using a transistor Q_2 connected in the CG configuration. Here V_{G2} is the dc bias voltage.



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7.3.2. **The MOS Cascode**

- Figure 7.7(a) shows the MOS cascode amplifier without a load circuit and with the gate of Q_2 connected to signal ground.
  - This circuit is valid for small-signal calculations only.
- Objective is to determine the parameters g_m and R_O of the equivalent circuit shown in Figure 7.7(b).
- If the node d_2 of the equivalent circuit is short-circuited to ground, the current flowing through the short-circuit will equal $G_m v_i$.
  - Note that $G_m = i_o/v_i$.

7.3.2. **The MOS Cascode**

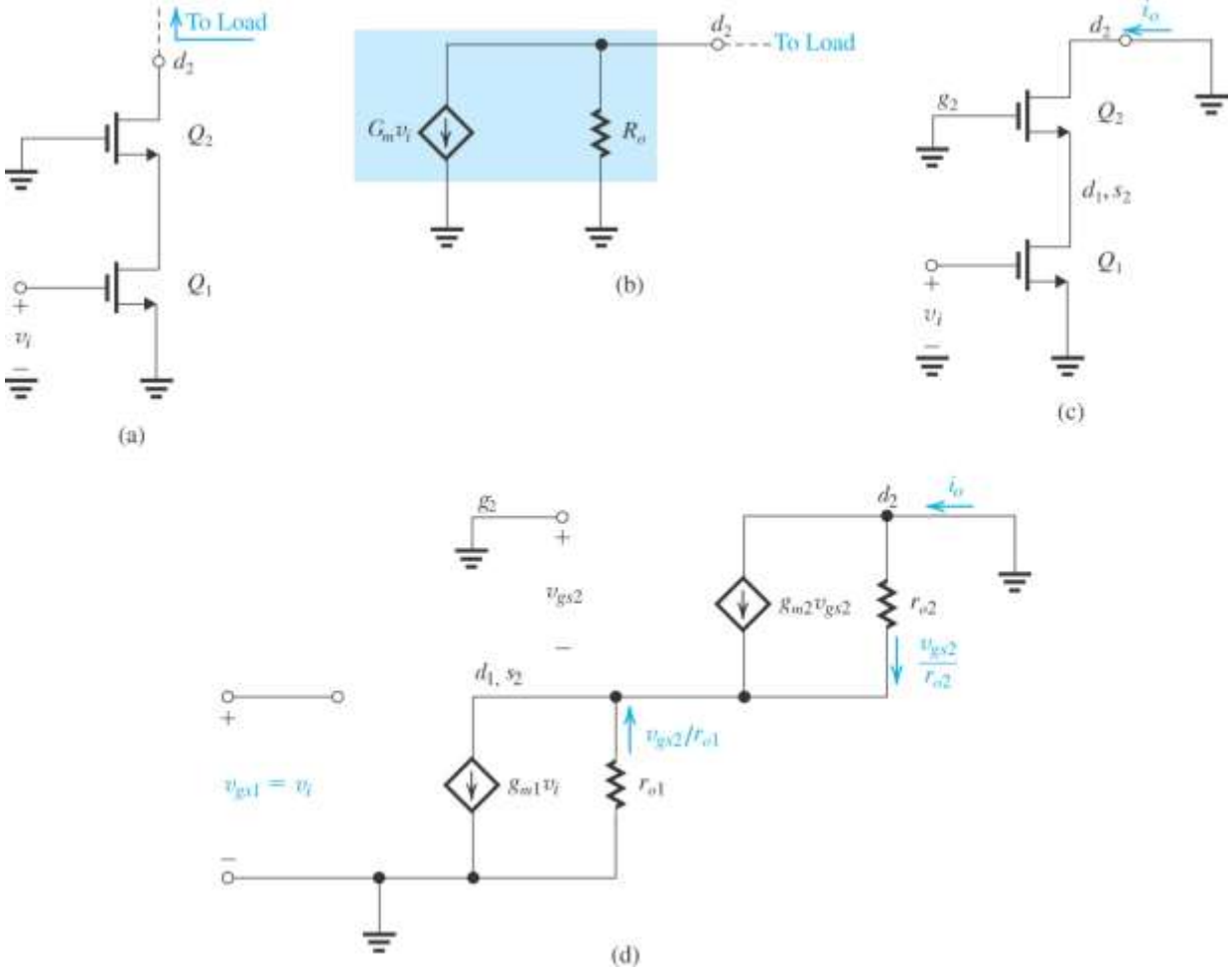


Figure 7.7: (a) A MOS cascode amplifier prepared for small-signal calculations; **(b)** output equivalent circuit of the amplifier in (a); **(c)** the cascode amplifier with the output short-circuited to determine G_m; **(d)** equivalent circuit of the situation in (c).

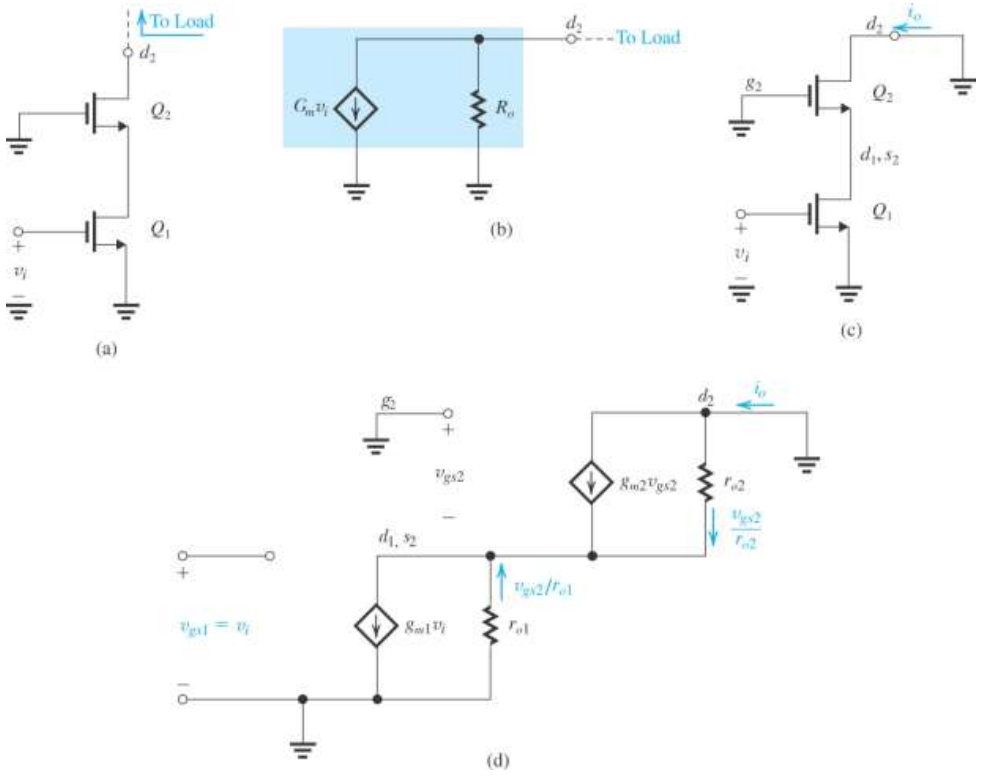
7.3.2. The MOS Cascode

for MOS Cascode Amplifier:

(7.21) $g_{m2}v_{gs2} = g_{m1}v_i$

(7.22) $G_m = \dfrac{i_o}{v_i} = g_{m1}$

(7.23) $-v_{gs2} = i_x r_{o1}$



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7.3.2. The MOS Cascode

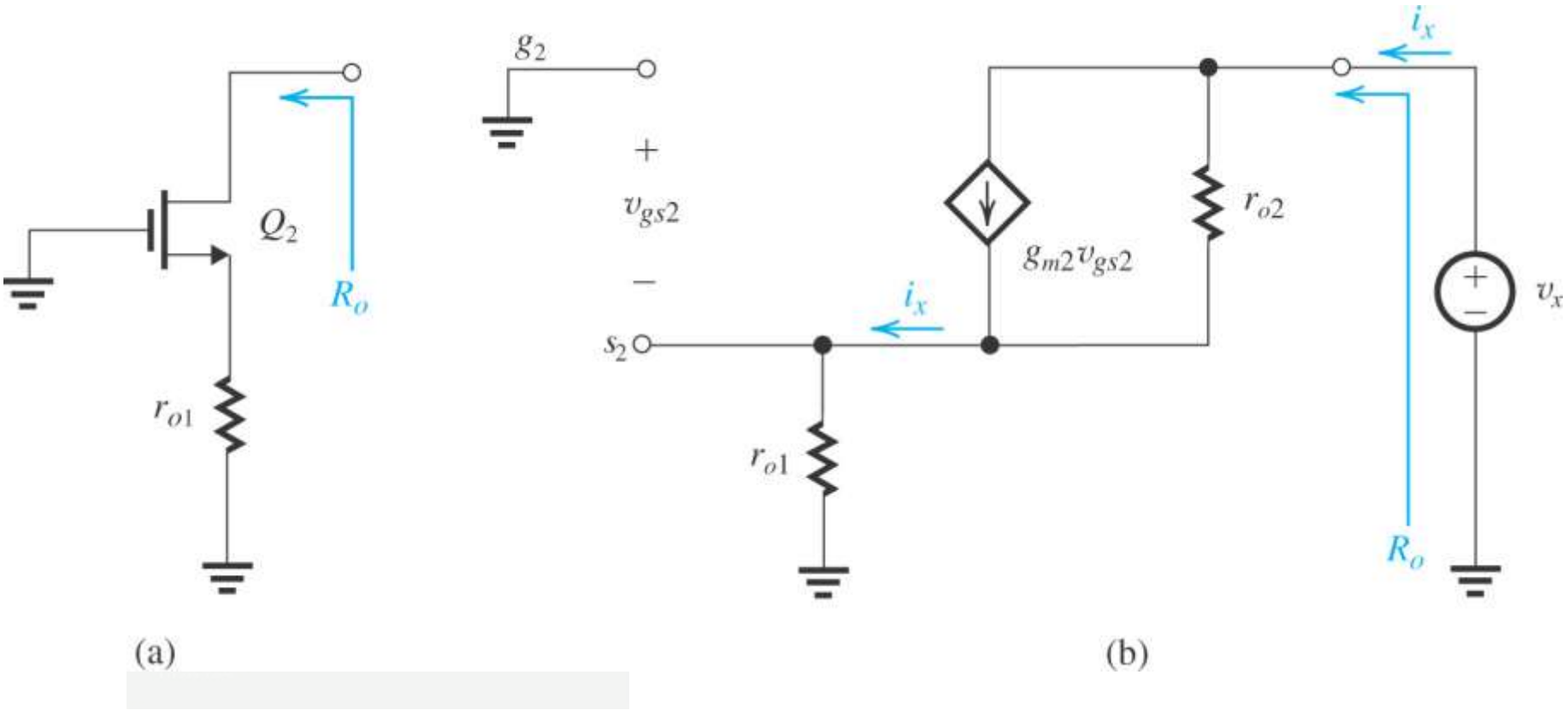


Figure 7.8: Determining the output resistance of the MOS cascode amplifier.

7.3.2. **The MOS Cascode**

- If the cascode amplifier is loaded with an ideal constant-current source as shown in Figure 7.9(a), the voltage gain realized can be found from the equivalent circuit in Figure 7.9(b) as $A_{vo} = -g_{m1}R_o$.

for MOS Cascode Amplifier:

$$(7.24)\ R_o = r_{o1} + r_{o2} + g_{m2}r_{o2}r_{o1}$$

$$(7.25)\ R_o = \left(g_{m2}r_{o2}\right)r_{o1}$$

$$(7.26)\ A_{vo} = \frac{v_o}{v_i} = -\left(g_{m1}r_{o1}\right)\left(g_{m2}r_{o2}\right)$$

$$(7.27)\ A_{vo} = -A_0^2$$

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7.3.2. **The MOS Cascode**

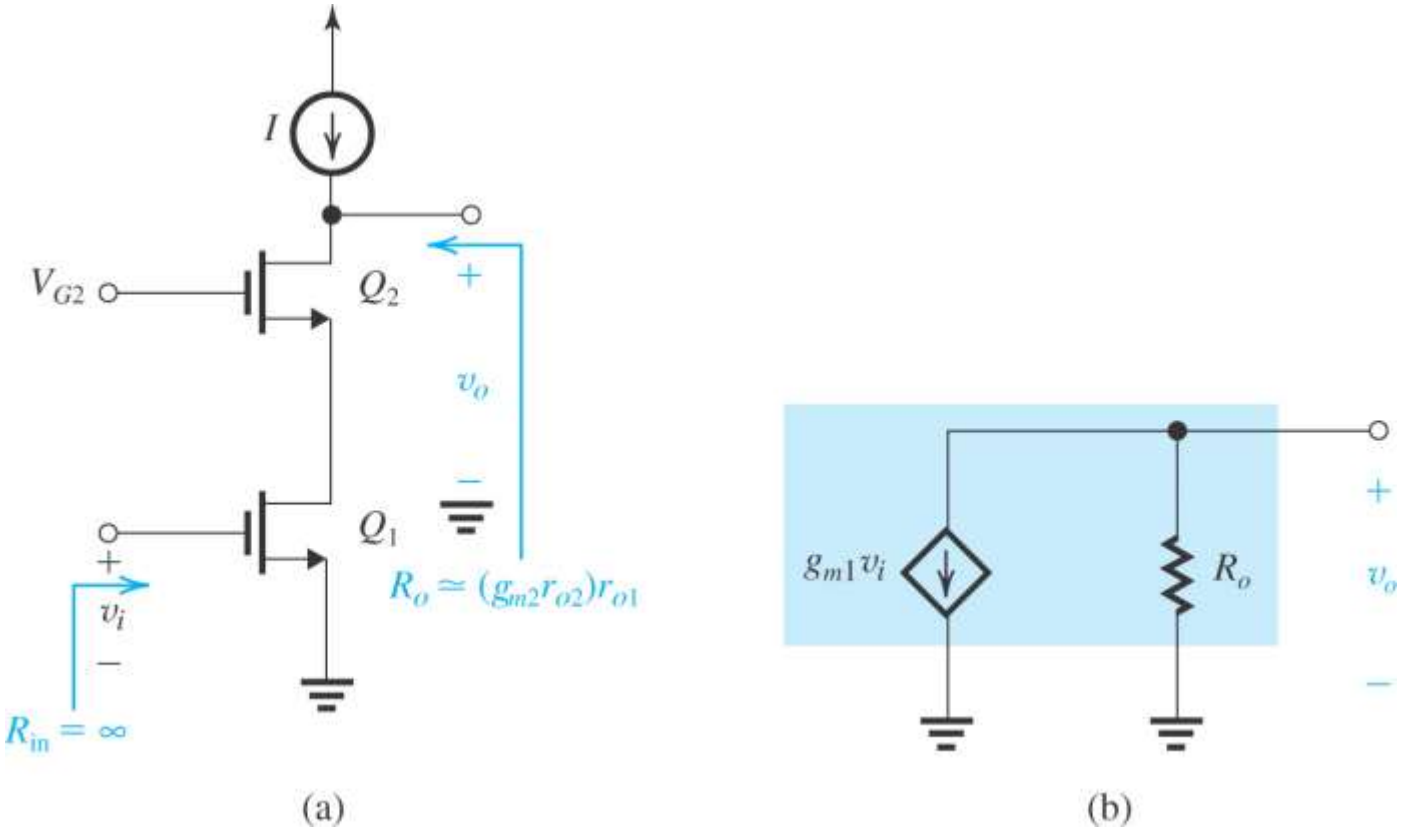


Figure 7.9 (a) A MOS cascode amplifier with an ideal current-source load; **(b)** equivalent circuit representation of the cascode output.

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7.3.2. The MOS Cascode

- Cascoding may be employed to raise the output resistance of the current-source load as shown in Figure 7.10.
  - Here, Q_4 is the current-source transistor and Q_3 is the CG cascode transistor.
  - Voltage V_{G3} and V_{G4} are the dc bias voltages.
  - The cascode transistor (Q_3) multiplies the output resistance of Q_4, r_{o4} to provide an output resistance for the cascode current source of…
    - $R_o = (g_{m3}r_{o3})r_{o4}$
- Combining a cascode amplifier with a cascode current source results in the circuit shown in Figure 7.11.

7.3.2. The MOS Cascode

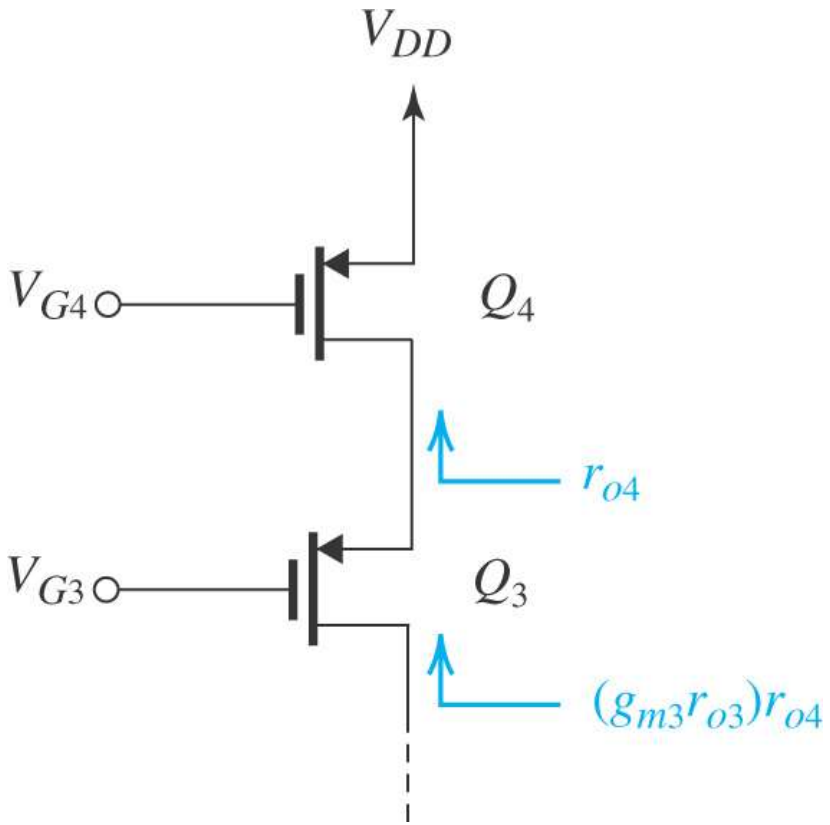


Figure 7.10 Employing a cascode transistor Q_3 to raise the output resistance of the current source Q_4.

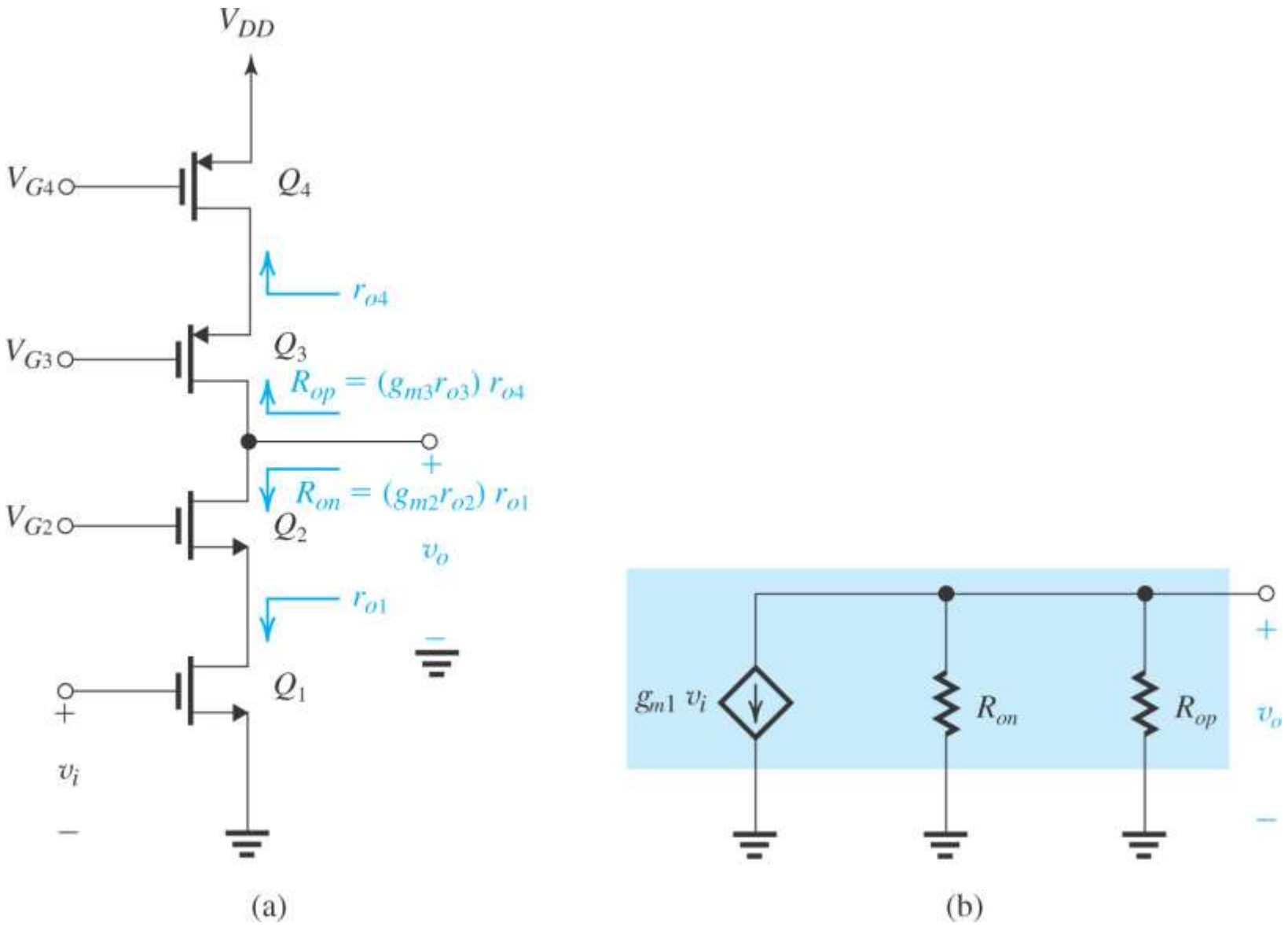


Figure 7.11: A cascode amplifier with a cascode current-source load.

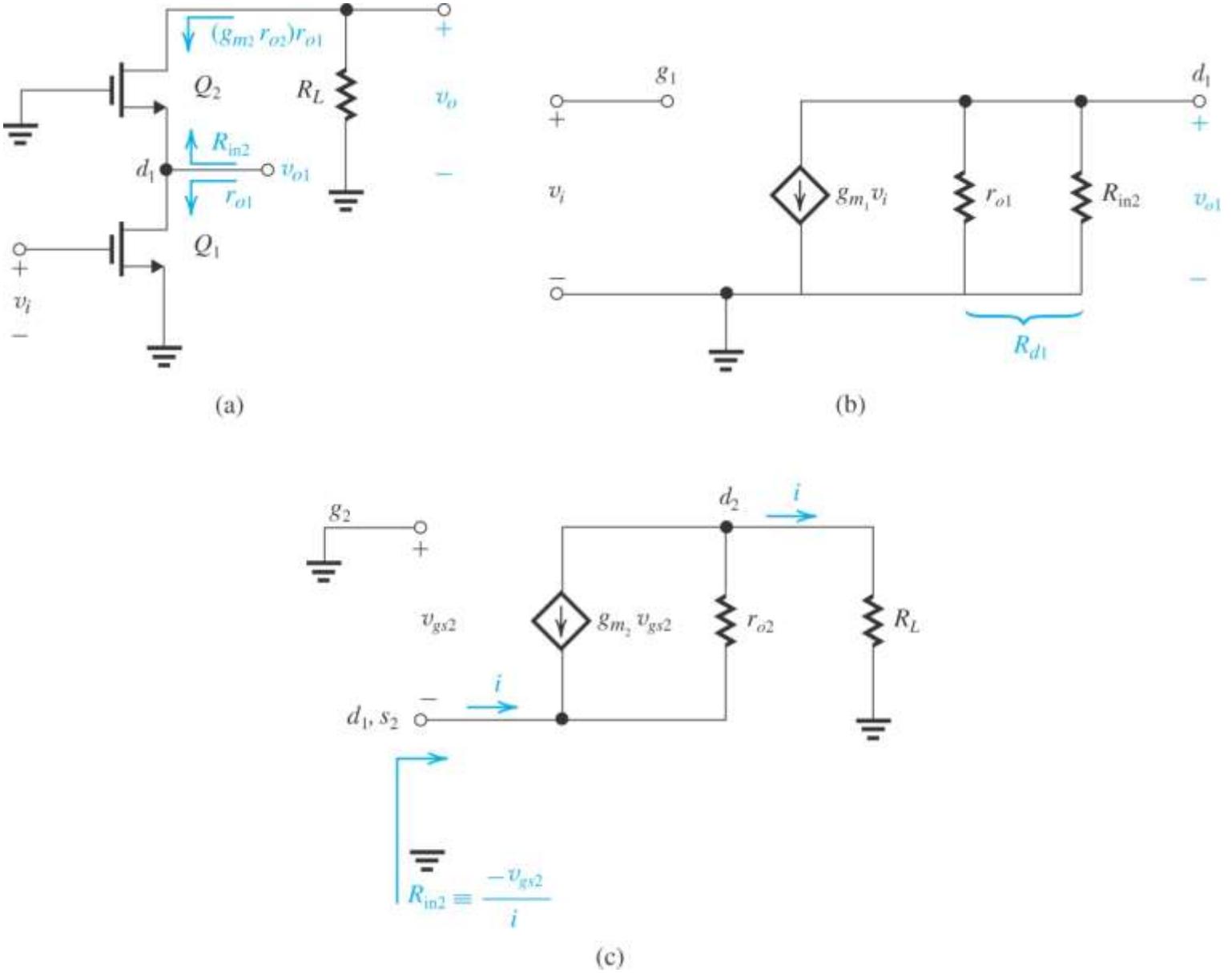


Figure 7.12: (a) The cascode amplifier with a load resistance R_L. Only signal quantities are shown. **(b)** Determining v_{01}. **(c)** Determining R_{in2}.

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7.3.3. Distribution of Voltage Gain in a Cascode Amplifier

$$(7.31)\ A_v = -g_{m1}\left(g_{m2}r_{o2}r_{o1}\,||\,R_L\right)$$

$$(7.32)\ A_v = A_{v1}A_{v2} = \left(\frac{v_{o1}}{v_i}\right)\left(\frac{v_o}{v_{o1}}\right)$$

$$(7.33)\ A_{v1} = \frac{v_{o1}}{v_i} = -g_{m1}R_{d1}$$

$$(7.34)\ R_{in2} = \frac{R_L + r_{o2}}{1 + g_{m2}r_{o2}}$$

$$(7.35)\ R_{in2} \approx \frac{R_L}{g_{m2}r_{o2}} + \frac{1}{g_{m2}}$$

7.3.3. Distribution of Voltage Gain in a Cascode Amplifier

- If r_{o2} is infinite, as was assumed previously, then R_{in2} reduces to $1/g_{m2}$ – verifying the result found there.
- If r_{o2} cannot be neglected, as is always the case in IC amplifiers, the input resistance depends on the value of R_L in an interesting fashion.
  - The load resistance (R_L) is divided by the factor ($g_{m2}r_{o2}$).
- This transformation is illustrated in Figure 7.13.

7.3.3. **Distribution of Voltage Gain in a Cascode Amplifier**

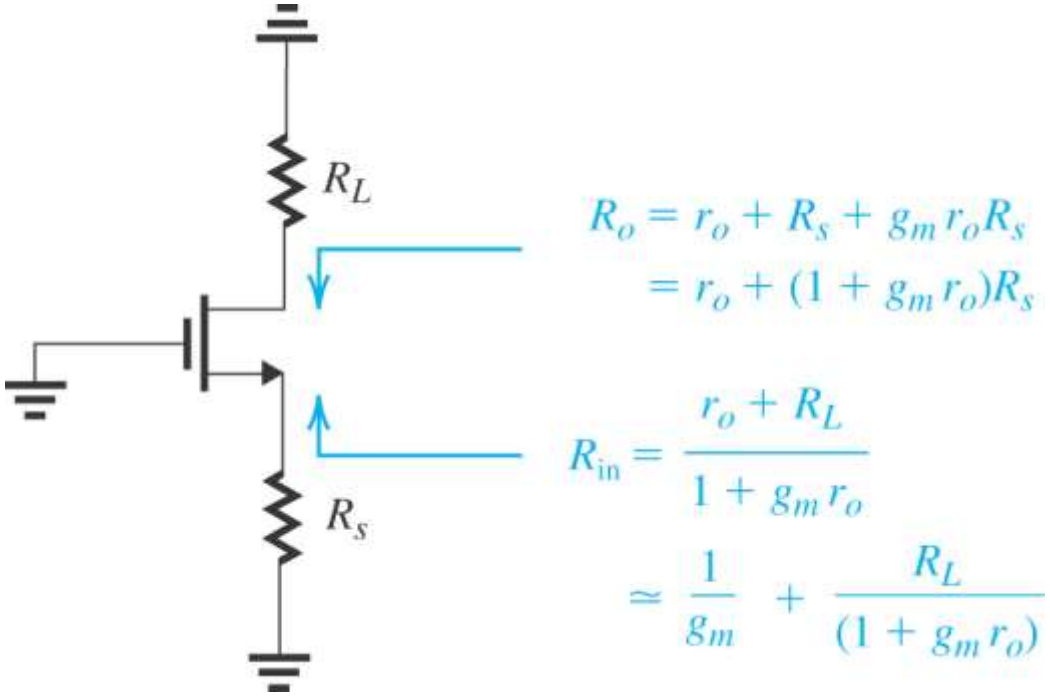


Figure 7.13: The impedance-transformation properties of the common-gate amplifier. Depending on the values of R_S and R_L, one can sometimes write $R_{in} = R_L/(g_m r_o)$ and $R_o = (g_m r_o)R_S$. However, such approximations are not always justified.

7.3.3. Distribution of Voltage Gain in a Cascode Amplifier

Table 7.1 Gain Distribution in the MOS Cascode Amplifier for Various Values of R_L

Case	R_L	R_{in2}	R_{d1}	A_{v1}	A_{v2}	A_v
1	∞	∞	r_o	$-g_m r_o$	$g_m r_o$	$-(g_m r_o)^2$
2	$(g_m r_o) r_o$	r_o	$r_o/2$	$-\frac{1}{2}(g_m r_o)$	$g_m r_o$	$-\frac{1}{2}(g_m r_o)^2$
3	r_o	$\frac{2}{g_m}$	$\frac{2}{g_m}$	-2	$\frac{1}{2}(g_m r_o)$	$-(g_m r_o)$
4	0	$\frac{1}{g_m}$	$\frac{1}{g_m}$	-1	0	0

7.3.4. The Output Resistance of a Source-Degenerated CS Amplifier

- Previous sections discuss benefits obtained when resistance R_S is included in the source lead of the CS amplifier.
  - Such a resistance is referred to as a **source-degeneration resistance** because of its action is reducing the effective transconductance of the CS stage to $g_m/(1+g_mR_s)$.
- Output resistance is defined as below.

$$(7.38)\ R_o = R_S + r_o + g_m r_o R_S$$

$$(7.39)\ R_o \approx \left(1 + g_m R_S\right) r_o$$

7.3.4. **The Output Resistance of a Source-Degenerated CS Amplifier**

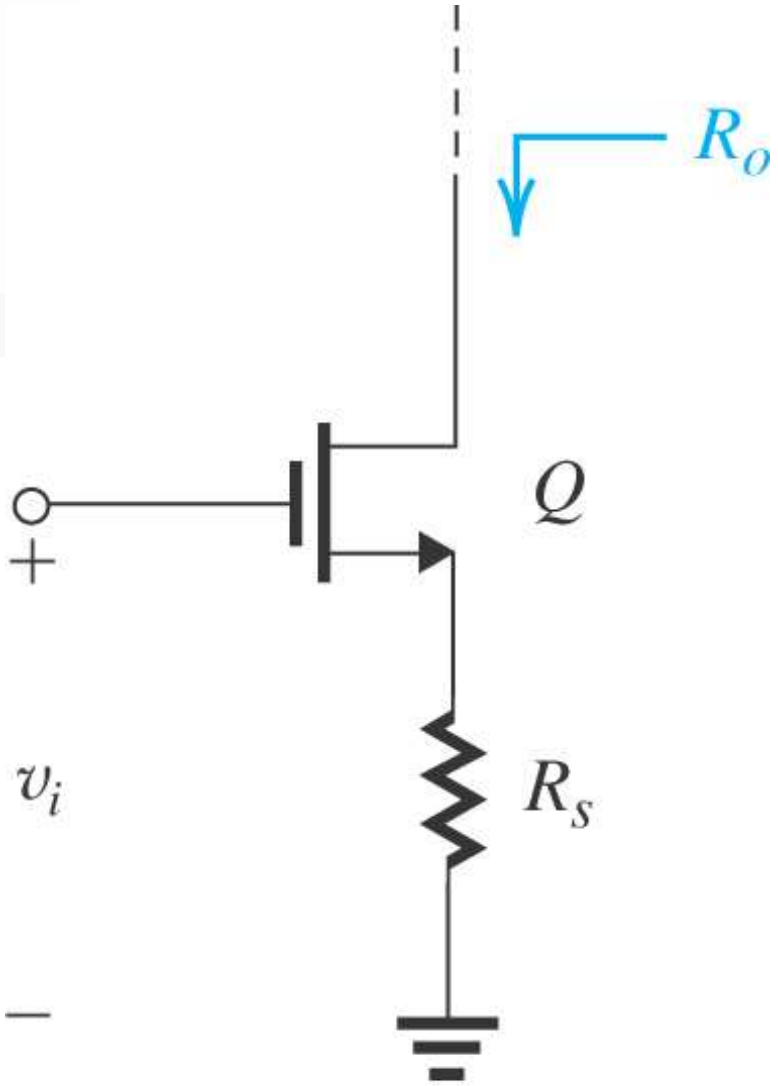


Figure 7.14: The output resistance expression of the cascode can be used to find the output resistance of a source-degenerated common-source amplifier. Here, a useful interpretation of the result is that R_s increases the output resistance by the factor $(1 + g_m R_s)$.

$$R_o = R_s + r_o + g_m r_o R_s$$
$$\simeq (1 + g_m R_s)\, r_o$$

7.3.5. **Double Cascoding**

- If still a higher output resistance and correspondingly higher gain are required, it is still possible to add another level of cascoding – as illustrated in Figure 7.15.
- Observe that Q_3 is the second cascode transistor, and it raises the output resistance by $(g_{m3}r_{o3})$.

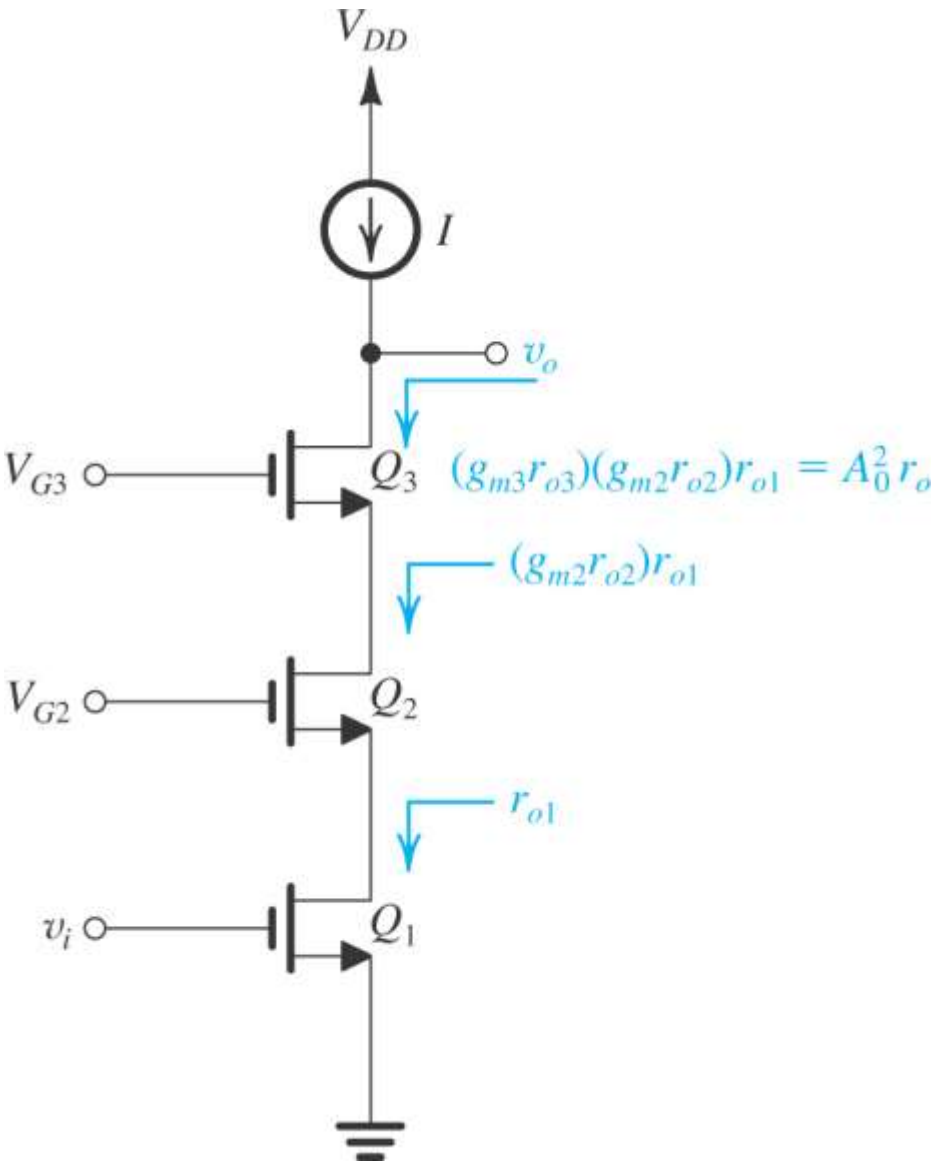


Figure 7.15: Double cascoding.

7.3.6. **The Folded Cascode**

- To avoid the problem of stacking a large number of transistors across a low-voltage power supply, one may use a PMOS transistor for the cascode device – as shown in Figure 7.16.
- This provides an alternative to the design proposed in previous section.

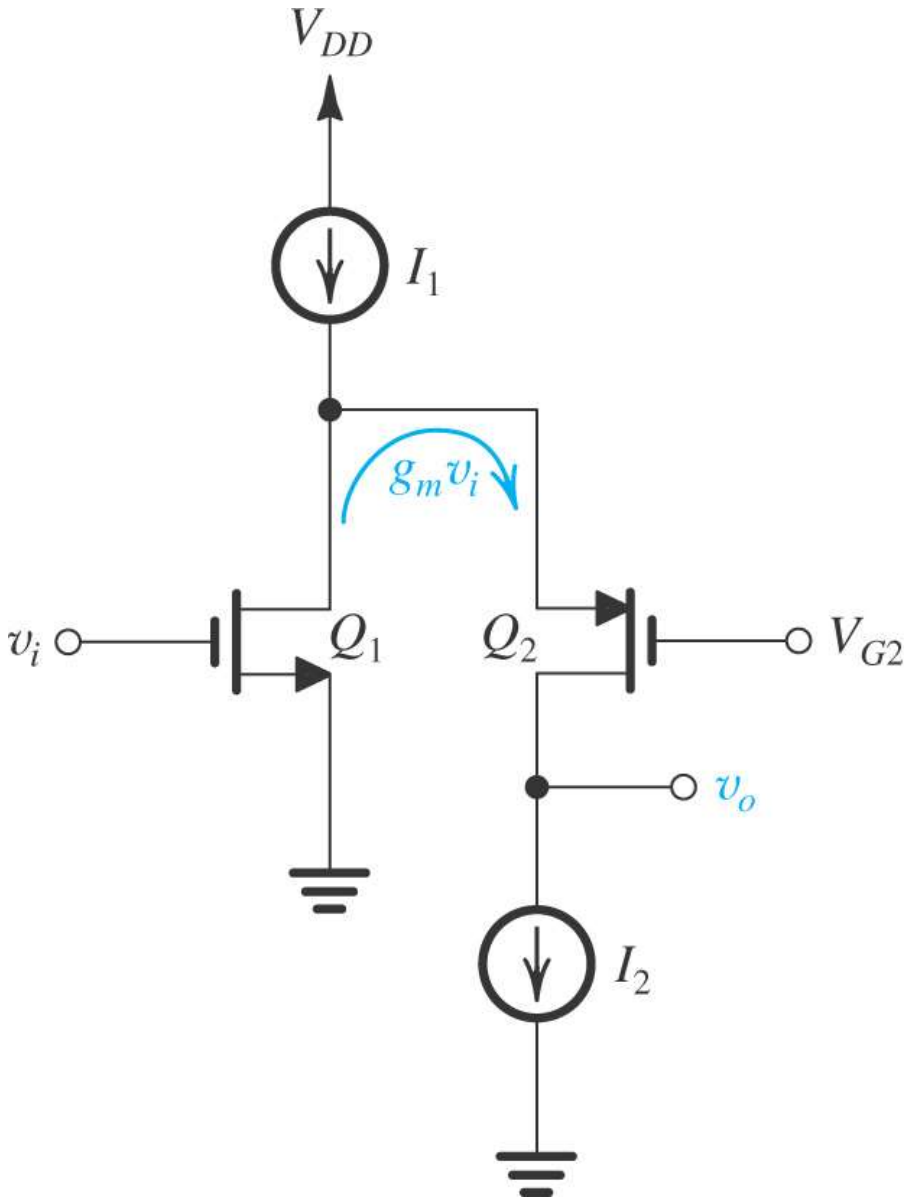


Figure 7.16: The folded cascode.

7.3.7. **The BJT Cascode**

- Figure 7.17(a) shows the BJT cascode amplifier with an ideal current-source load.
- Voltage V_{B2} is a dc bias voltage for the CB cascode transistor Q_3.
- Objective is to determine the parameters G_m and R_o of the equivalent circuit of Figure 7.17(b).
- As in case of MOS cascode, G_m is the short-circuit transconductance and can be determined from the circuit in Figure 7.17(c).

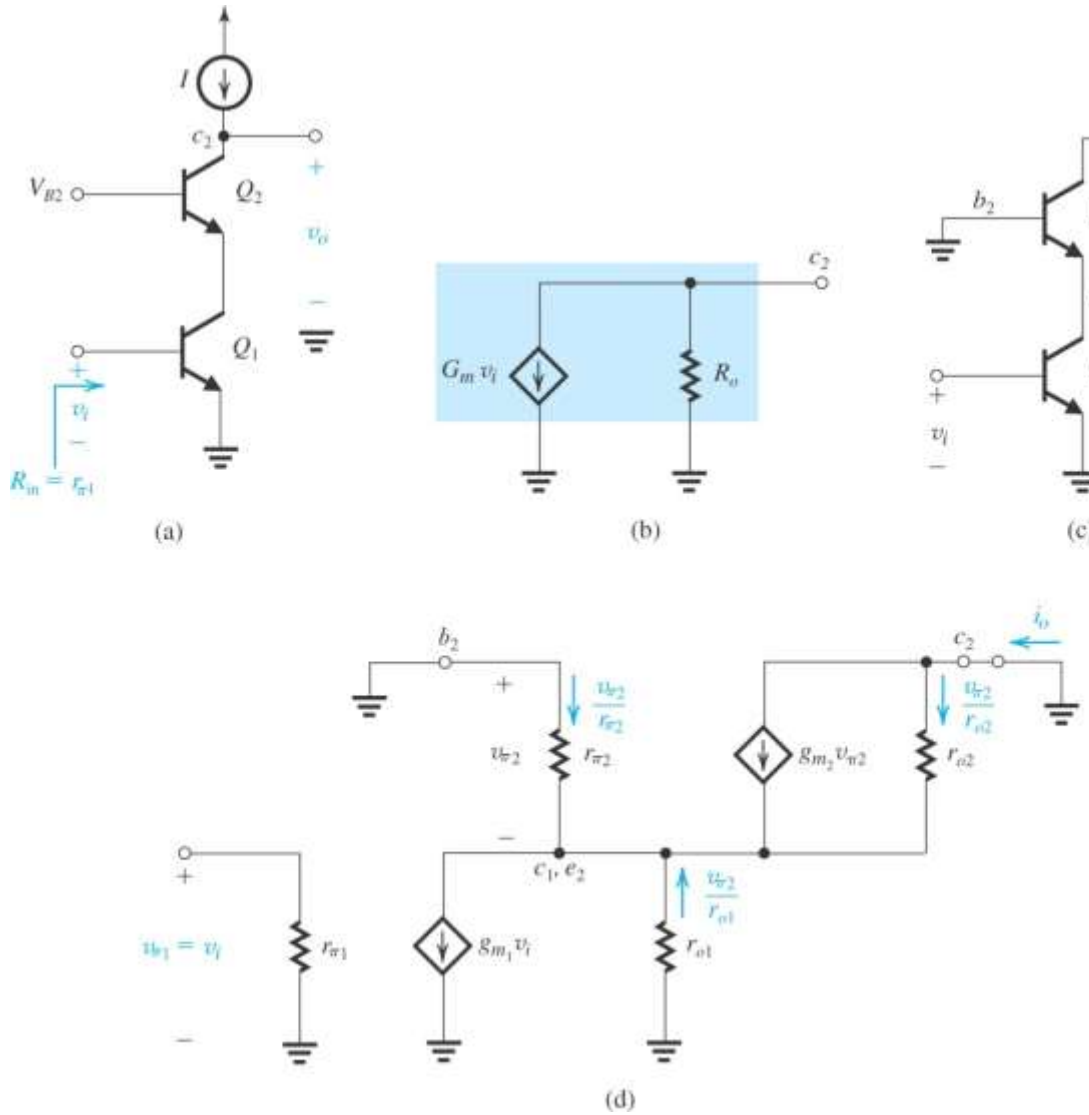


Figure 7.17: (a) A BJT cascode amplifier with an ideal current-source load; **(b)** small-signal equivalent-circuit representation of the output of the cascode amplifier; **(c)** the cascode amplifier with the output short-circuited to ground, and **(d)** equivalent circuit representation of **(c)**.

7.3.7. **The BJT Cascode**

$$(7.41)\ g_{m2}v_{\pi 2} \approx g_{m1}v_i$$

$$(7.42)\ -v_{\pi 2} = i_x\left(r_{o1}\,||\,r_{\pi 2}\right)$$

$$(7.43)\ R_o = r_{o2} + \left(r_{o1}\,||\,r_{\pi 2}\right) + \left(g_{m2}r_{o2}\right)v_{\pi 2}$$

$$(7.44)\ R_o \approx r_{o2} + \left(g_{m2}r_{o2}\right)\left(r_{o1}\,||\,r_{\pi 2}\right)$$

$$(7.45)\ R_o \approx \left(g_{m2}r_{o2}\right)\left(r_{o1}\,||\,r_{\pi 2}\right)$$

$$(7.46)\ R_o\big|_{\max} \approx g_{m2}r_{\pi 2}\,r_{o2}$$

β

7.3.7. **The BJT Cascode**

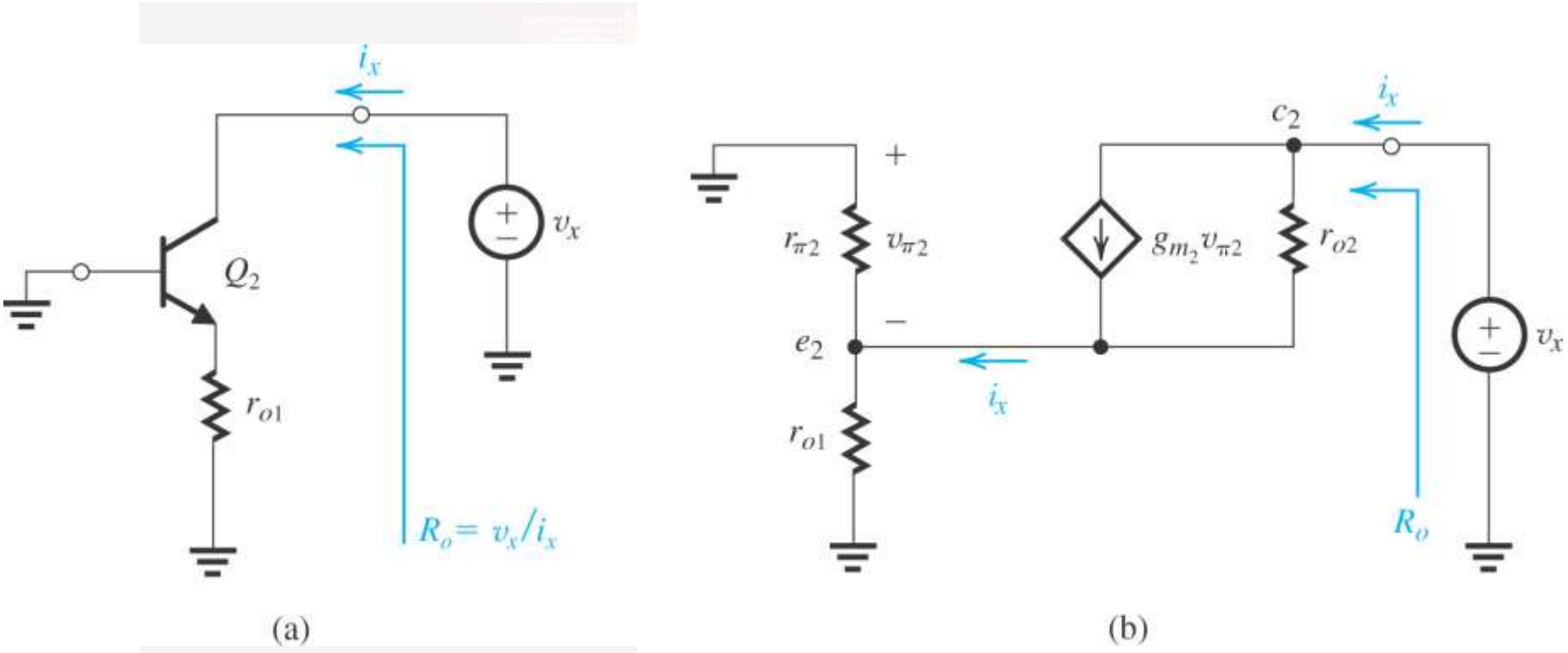


Figure 7.18: Determining the output resistant R_o of the BJT cascode amplifier.

7.3.7. **The BJT Cascode**

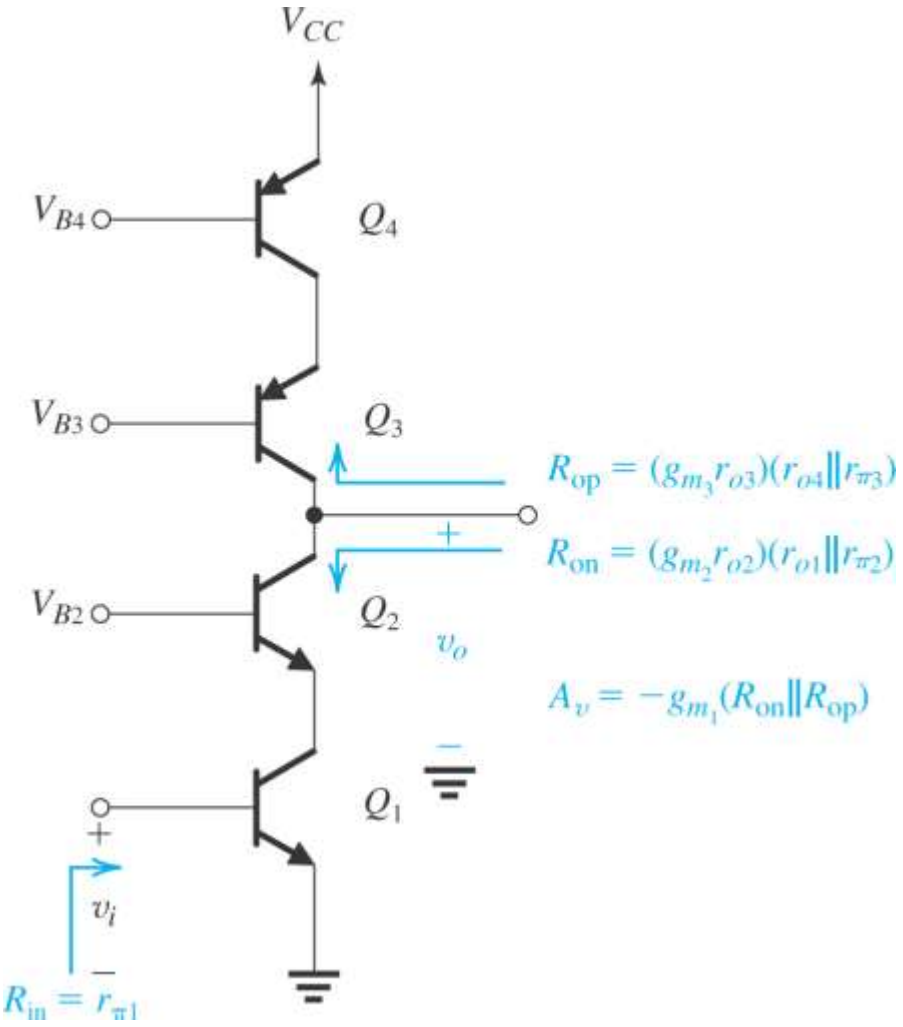


Figure 7.19: Determining the output resistant R_o of the BJT cascode amplifier.

7.3.8. The Output Resistance of an Emitter-Degenerated CE Amplifier

- As done in MOS case, one may adapt the expression for R_o derived from BJT cascode (equation 7.43).
- For the case of a CE amplifier with resistance R_e connected in its emitter – as shown in Figure 7.20(a).
- The output resistance is obtained from equation 7.43 by replacing r_{o2} with r_o, g_{m2} with g_m, $r_{\pi 2}$ with r_π, and r_{o1} with R_e.

$$(7.50)\ R_o = r_o + (R_e \,||\, r_\pi) + (g_m r_o)(R_e \,||\, r_\pi)$$

$$(7.51)\ R_o = \left[1 + g_m (R_e \,||\, r_\pi)\right] r_o$$

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7.3.8. The Output Resistance of an Emitter-Degenerated CE Amplifier

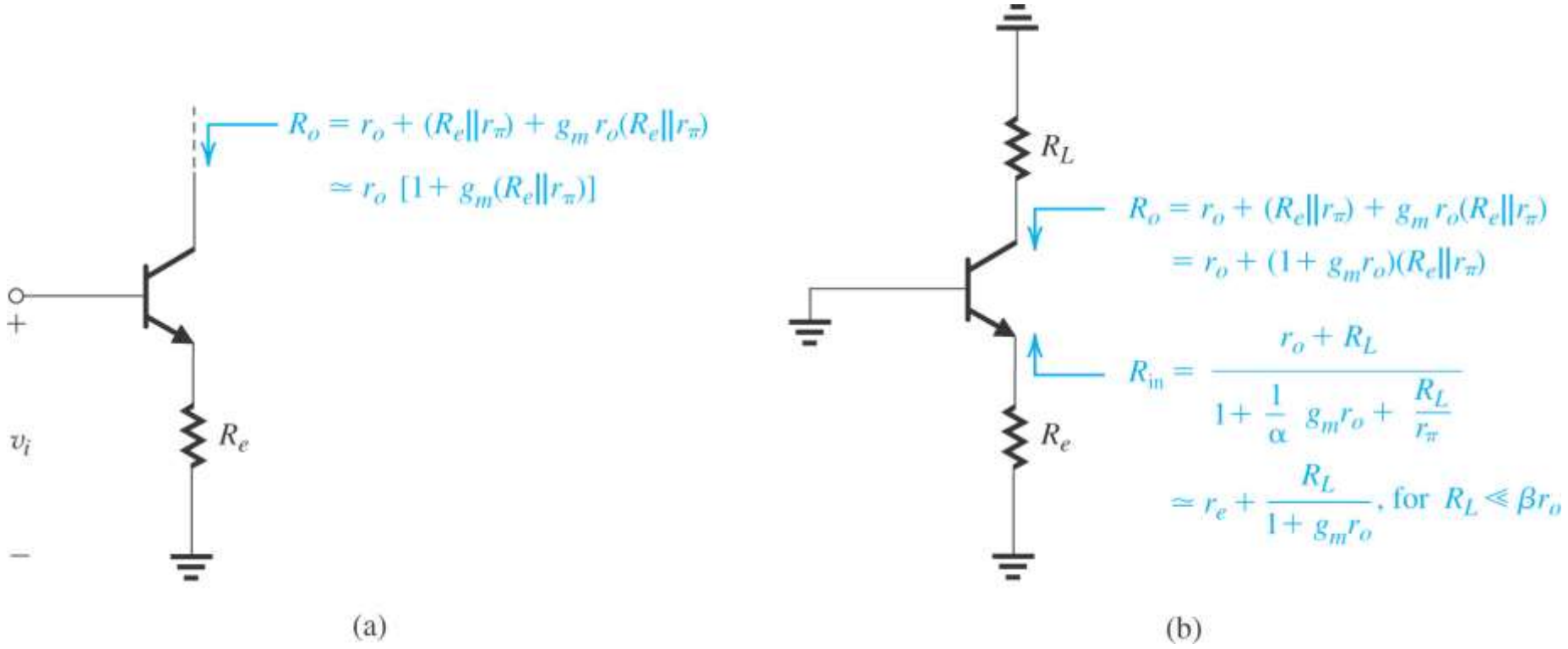


Figure 7.20: (a) Output resistance of a CE amplifier with emitter degeneration; **(b)** The impedance transformation properties of the CB amplifier. Note that for β = infinity, these formulas reduce to those for the MOSFET case (Figure 7.13).

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7.3.9. **BiMOS Cascodes**

- Certain advanced CMOS technologies allow the fabrication of bipolar transistors.
- They permit the circuit designer to combine MOS and bipolar transistors in circuits that take advantage of the unique features of each.
- The resulting technology is called BiCMOS.
- Figure 7.21. shows two possible BiCMOS cascode amplifiers.
  - The circuit in Figure 7.21(a) uses a MOS transistor for the amplifying device and a BJT for the cascode device.
  - The advantage of this circuit is an infinite input resistance as compared with all BJT case.

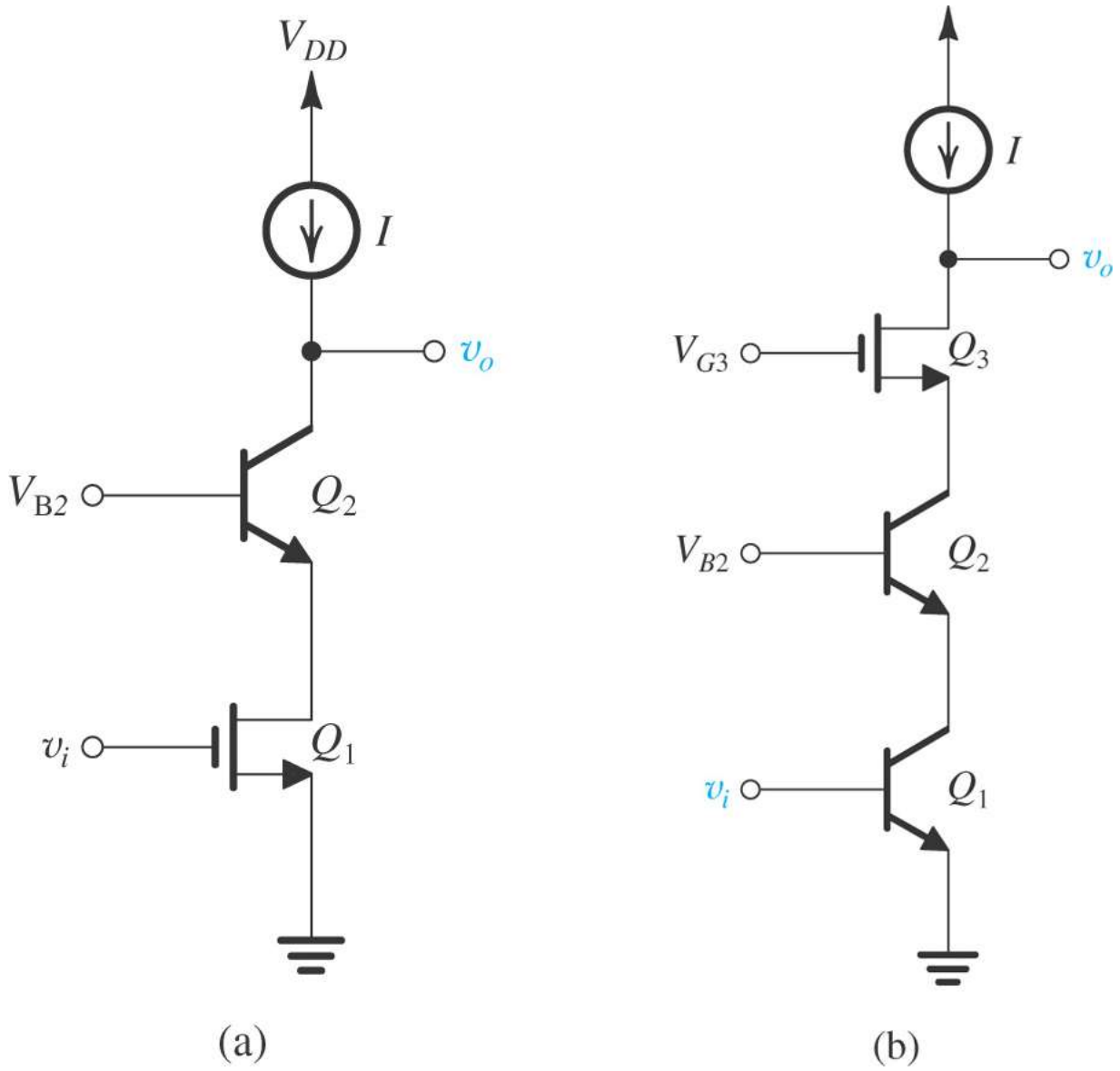


Figure 7.21: BiCMOS cascodes.

7.4. IC Biasing – Current Sources, Current Mirrors, and Current-Steering Circuits

- Biasing in integrated-circuit design is based on the use of constant-current sources.
  - On an IC chip with a number of amplifier stages, a constant dc current (**reference current**) is generated at one location and is then replicated at various other locations for biasing.
    - This is known as **current steering**.
- This approach has the advantage that the effort expended on generating a predictable and stable reference current need not be repeated.

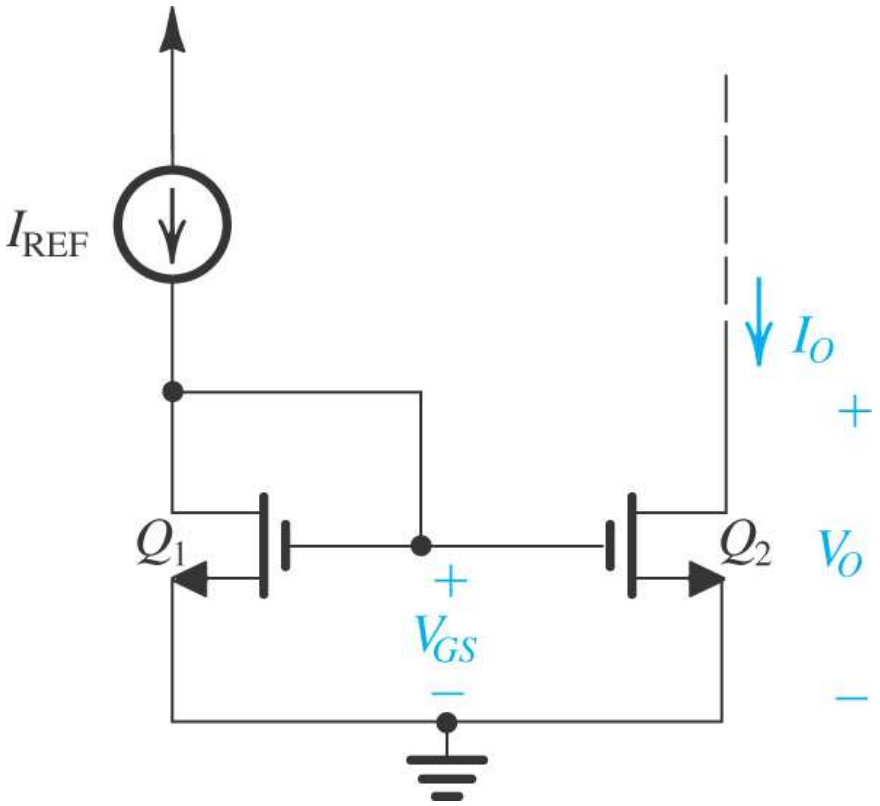
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7.4.1. The Basic MOSFET Current Source

- Figure 7.22. shows the circuit of a simple MOS constant-current source.
- The head of the circuit is transistor Q_1, the drain of which is shorted to the gate, thereby forcing it to operate in saturation mode – equation (7.52).
- The drain current of Q_1 is supplied by V_{DD} through resistor R which in most cases is located outside of the IC chip.
- If one considers transistor Q_2, it is realized that it has V_{GS} identical to Q_1 – thus (7.54) through (7.59) apply.

7.4.1. The Basic MOSFET Current Source

Figure 7.22 Circuit for a basic MOSFET constant-current source. For proper operation, the output terminal, that is, the drain of Q_2, must be connected to a circuit that ensures that Q_2 operates in saturation.



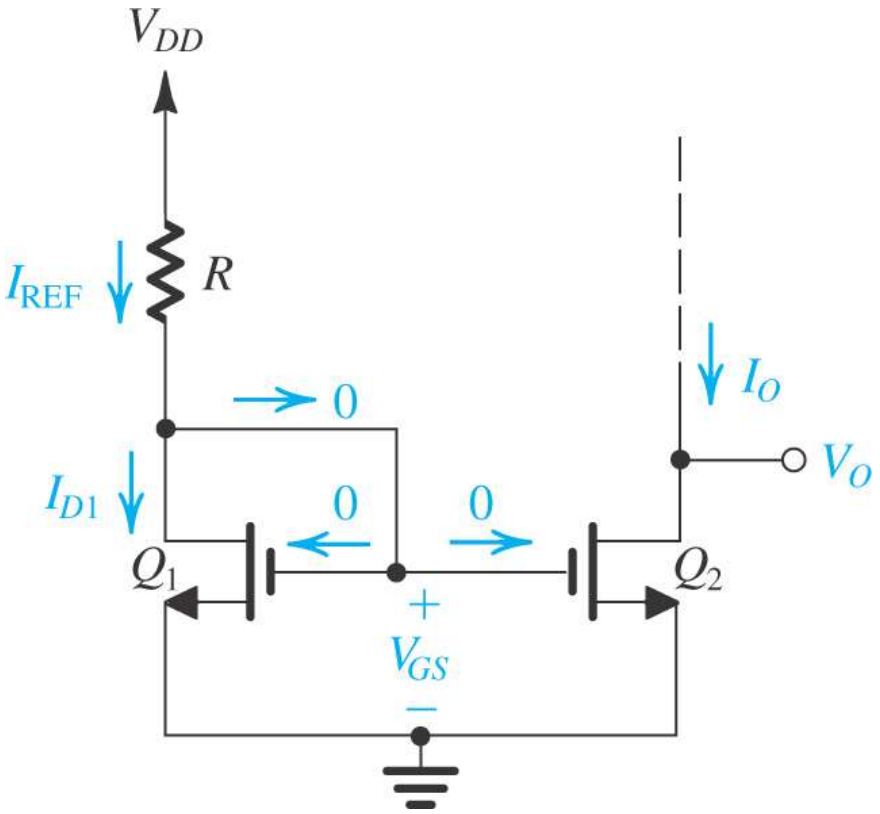


Figure 7.23: Basic MOSFET current mirror.

7.4.1. The Basic MOSFET Current Source

$$(7.52)\ I_{D1} = \frac{1}{2} k_n' \left(\frac{W}{L} \right)_1 \left(V_{GS} - V_{tn} \right)^2$$

$$(7.53)\ I_{D1} = I_{REF} = \frac{V_{DD} - V_{GS}}{R}$$

$$(7.54)\ I_O = I_{D2} = \frac{1}{2} k_n' \left(\frac{W}{L} \right)_2 \left(V_{GS} - V_{tn} \right)^2$$

$$(7.55)\ \frac{I_O}{I_{REF}} = \frac{(W/L)_2}{(W/L)_1}$$

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7.4.1. The Basic MOSFET Current Source

- The combination of Q_1 and Q_2 is referred to as a **current mirror**.
- Figure 7.23 depicts the current-mirror circuit with the input reference current shown as being supplied by a current source for both simplicity and generality.
- The **current gain** or **current transfer ratio** f the mirror is given by (7.55).
- The effect of V_o on I_o is defined in equations (7.56) through (7.59).

7.4.1. The Basic MOSFET Current Source

$$(7.56)\ V_O \geq V_{GS} - V_{tn}$$

$$(7.57)\ V_O \geq V_{OV}$$

$$(7.58)\ R_o \equiv \frac{\Delta V_O}{\Delta I_O} = r_{o2} = \frac{V_{A2}}{I_O}$$

$$(7.58)\ I_o = \frac{(W/L)_2}{(W/L)_1} I_{REF} \left(1 + \frac{V_O - V_{GS}}{V_{A2}} \right)$$

7.4.1. The Basic MOSFET Current Source

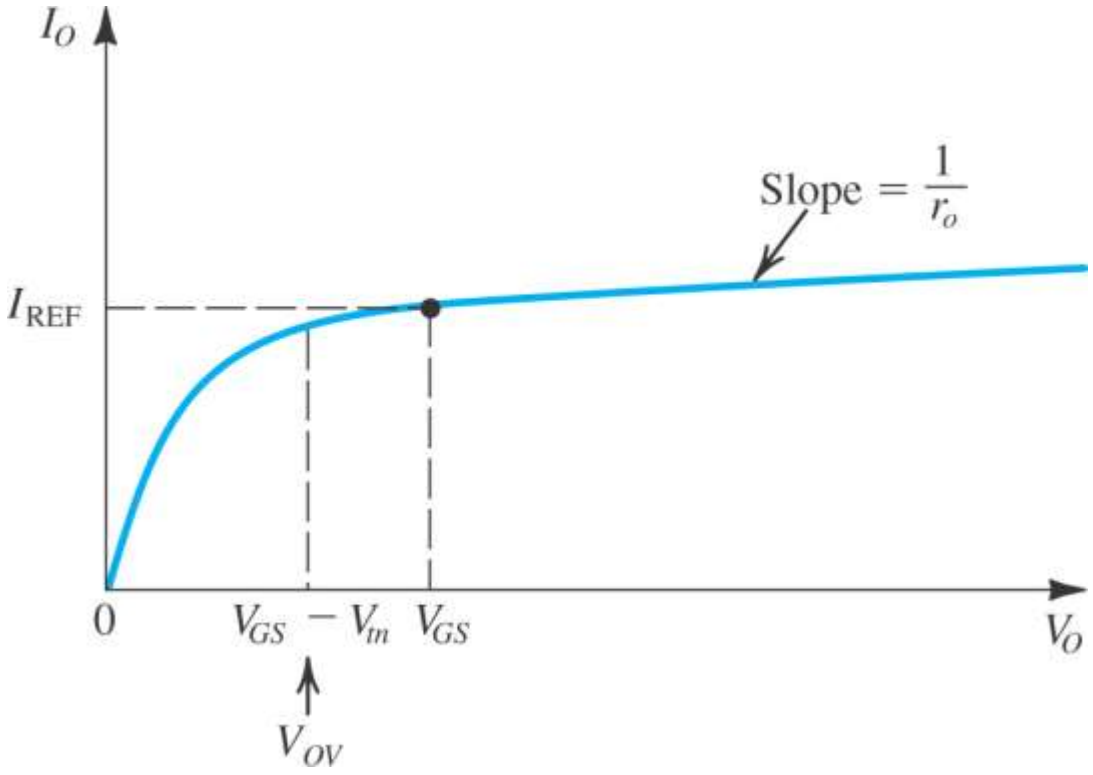


Figure 7.24: Output characteristic of the current source in Fig. 7.22 and the current mirror of Fig. 7.23 for the case of Q_2 matched to Q_1.

7.4.2. MOS Current-Steering Circuits

- Once a constant current has been generated, it can be replicated to provide dc bias or load current for the various stages of the amplifier in an IC.
- Current mirrors can be used to achieve this goal.

$$(7.60)\ I_2 = I_{REF}\frac{(W/L)_2}{(W/L)_1}$$

$$(7.61)\ I_3 = I_{REF}\frac{(W/L)_3}{(W/L)_1}$$

$$(7.62)\ V_{D2}, V_{D3} \geq -V_{SS} + V_{GS1} - V_{tn}$$

$$(7.63)\ V_{D2}, V_{D3} \geq -V_{SS} + V_{OV1}$$

$$(7.64)\ I_5 = I_4\frac{(W/L)_5}{(W/L)_4}$$

$$(7.65)\ V_{D5} = V_{DD} - |V_{OV5}|$$

7.4.2. MOS Current-Steering Circuits

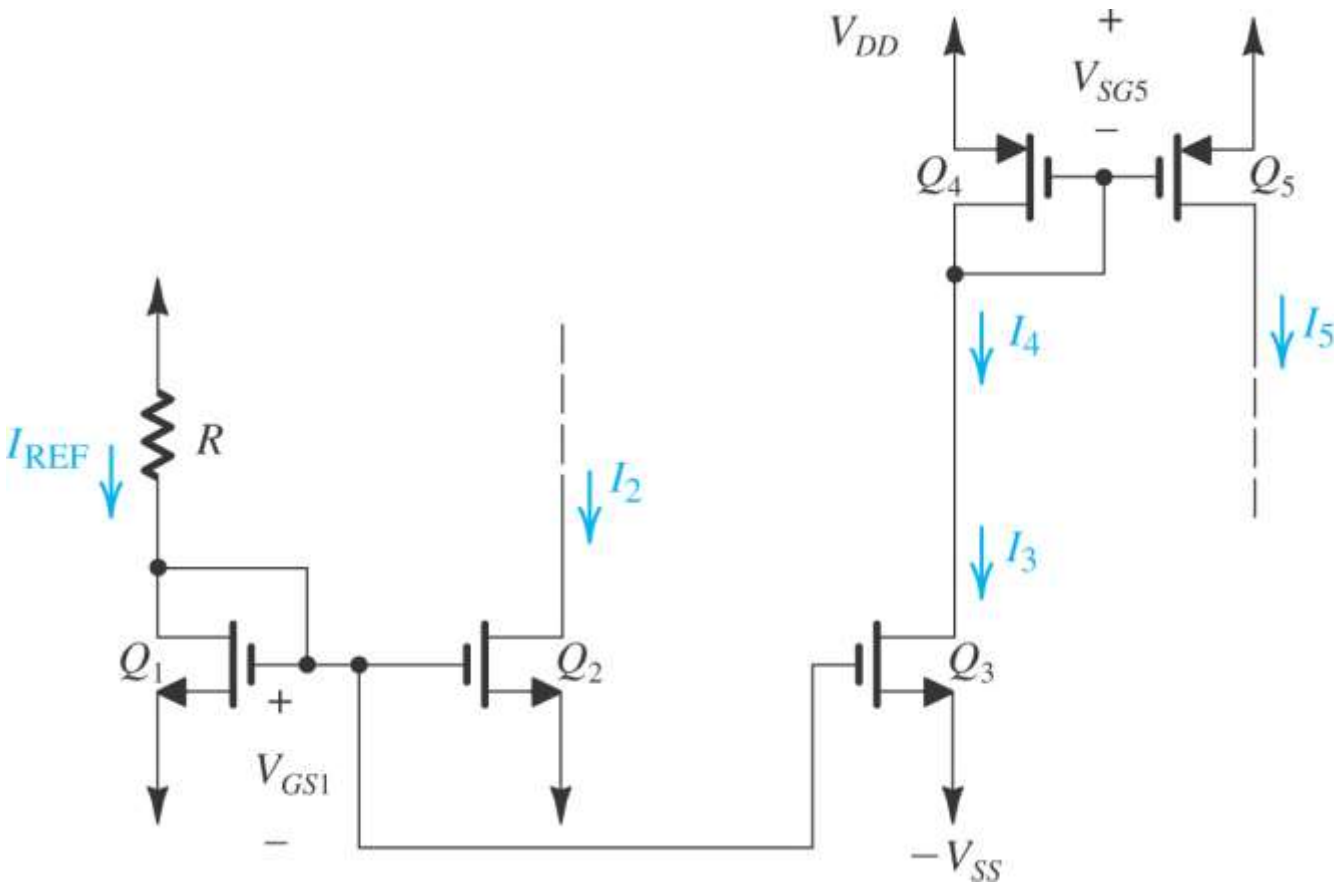


Figure 7.25: A current-steering circuit.

7.4.2. MOS Current-Steering Circuits

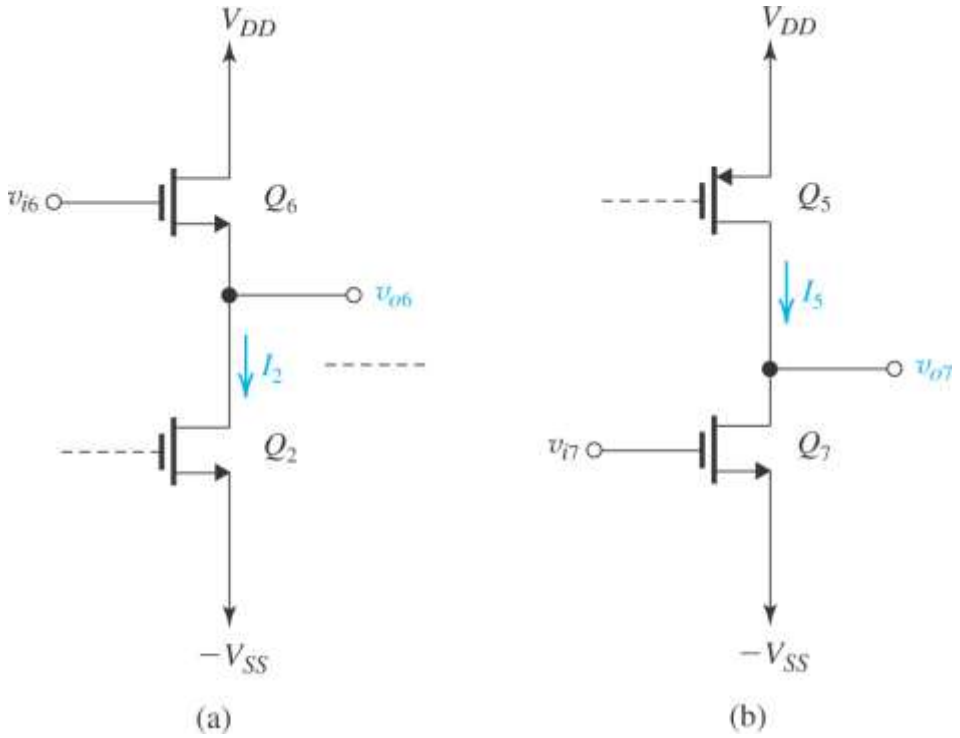


Figure 7.26: Application of the constant currents *I*2 and *I*5 generated in the current-steering circuit of Fig. 7.25. Constant-current *I2* is the bias current for the source follower *Q6*, and constant-current *I5* is the load current for the common-source amplifier *Q7*.

7.4.3. **BJT Circuits**

- The basic BJT current mirror is shown in Figure 7.28.
- It works in a fashion very similar to the MOS mirror. However, with two important differences:
  - The non-zero bias current causes an error in current mirroring (magnitude of current conducted).
  - The current transfer ratio is determined by the relative areas of the emitter-based junctions of Q_1 and Q_2.
- The textbook provides a more detailed description of the BJT-based current mirror – including equations which correspond to those provided for MOS-based mirror.

7.4.3. **BJT Circuits**

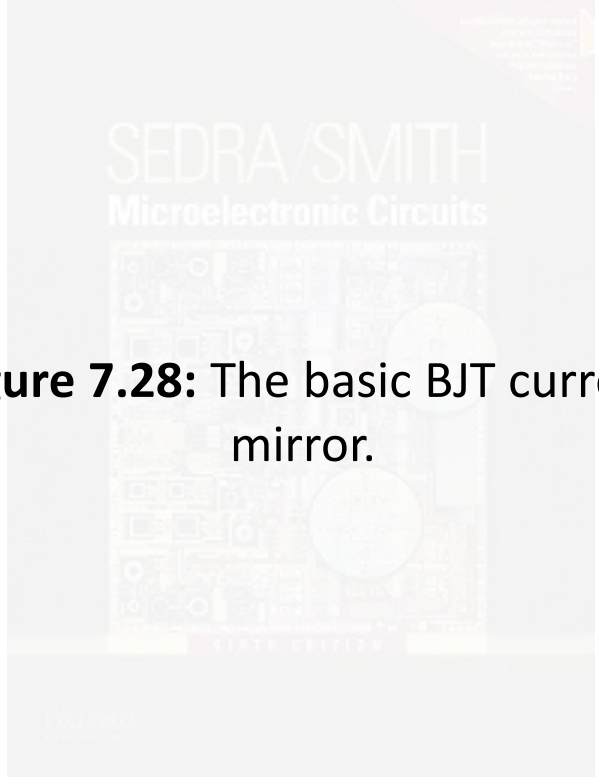
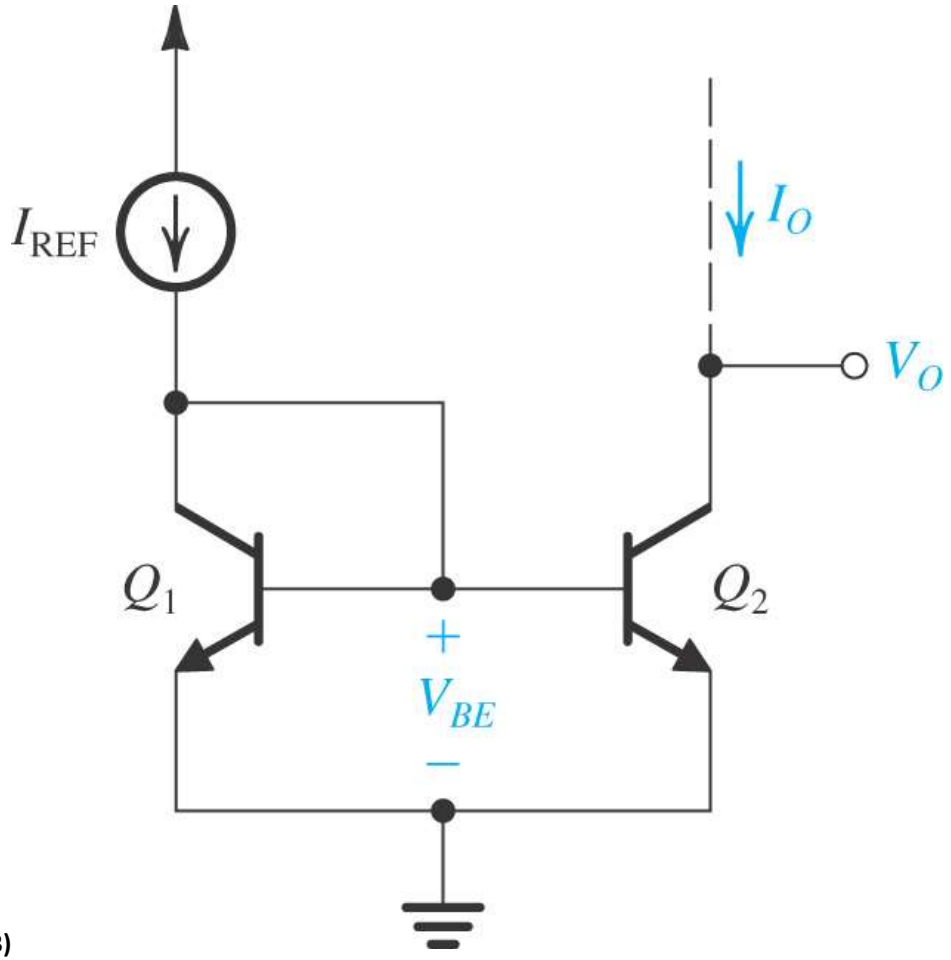


Figure 7.28: The basic BJT current mirror.



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7.5. Current-Mirror Circuits with Improved Performance

- Adaptations of the traditional current exist with improved performance. They include:
  - **7.5.1. Cascode MOS Mirror**
    - Previous sections demonstrate the cascoding of transistors may be used to increase gain and acquire “better” performance.
  - **7.5.2. A Bipolar Mirror with Base-Current Compensation**
    - Base-current compensation may be used to eliminate the effect of bias current on mirror operation. In other words, how may its operation be made more like the MOS implementation.

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7.5. Current-Mirror Circuits with Improved Performance

- Adaptations of the traditional current exist with improved performance. They include:
  - **7.5.3. The Wilson Current Mirror**
    - The addition of a diode-connected transistor in series with Q_2 may reduce the effect of β on output resistance.
  - **7.5.4. The Wilson MOS Mirror**
    - No β parameter exists for MOS. However, Wilson's adaptation may be used to increase output resistance and, in turn, gain.
  - **7.5.5. The Wildar Current Source**
    - A resistor R_E is included in the emitter lead of Q_2.

7.5. **Current-Mirror Circuits with Improved Performance**

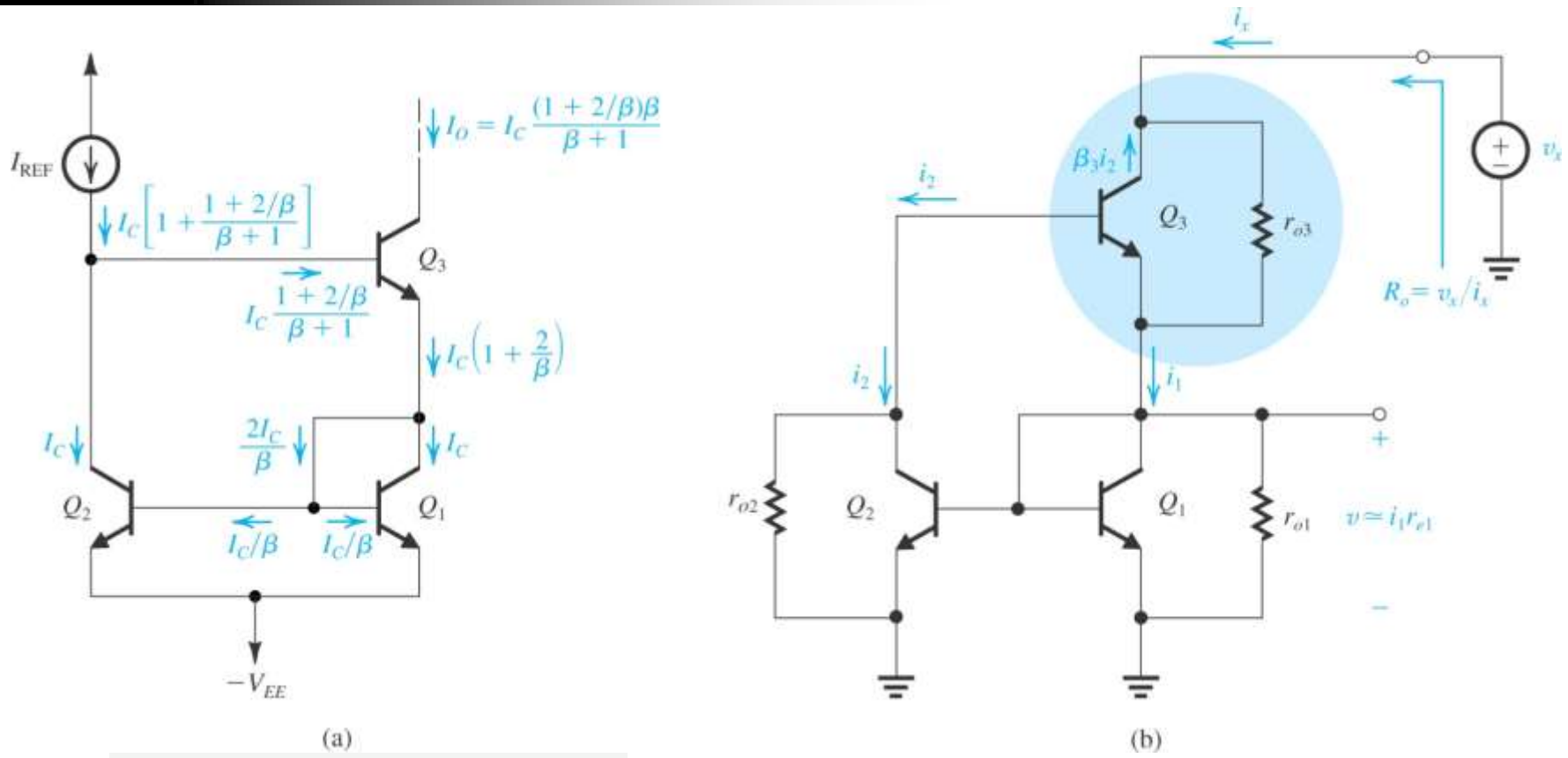


Figure 7.34: The Wilson bipolar current mirror: **(a)** circuit showing analysis to determine the current transfer ratio; **(b)** determining the output resistance.

7.5. **Current-Mirror Circuits with Improved Performance**

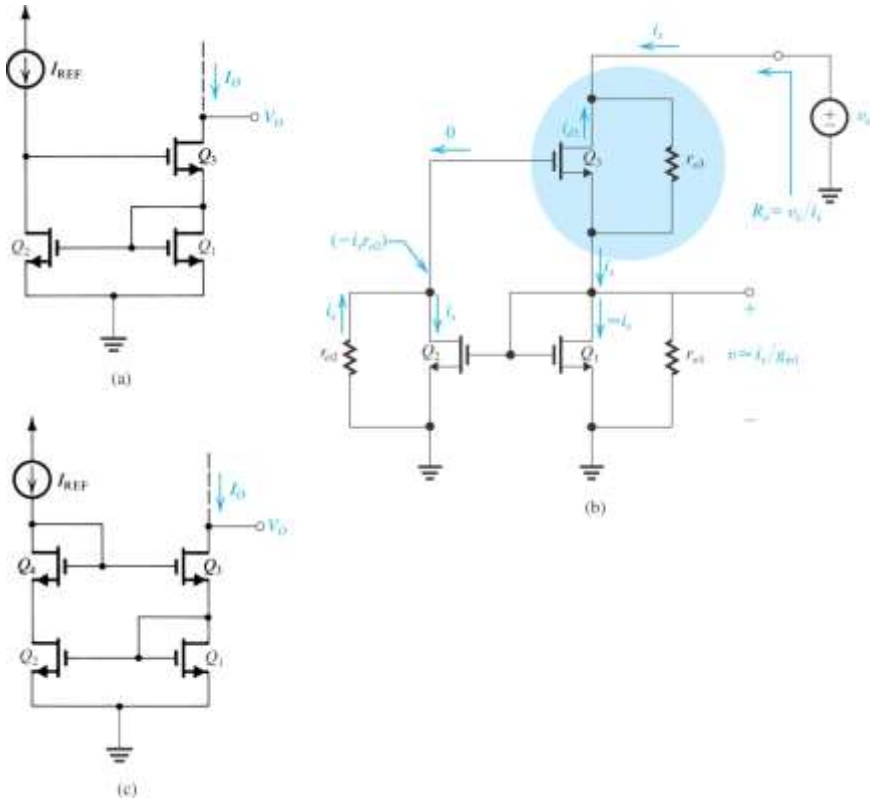


Figure 7.35: The Wilson MOS mirror: **(a)** circuit; **(b)** analysis to determine output resistance; **(c)** modified circuit.

7.5. **Current-Mirror Circuits with Improved Performance**

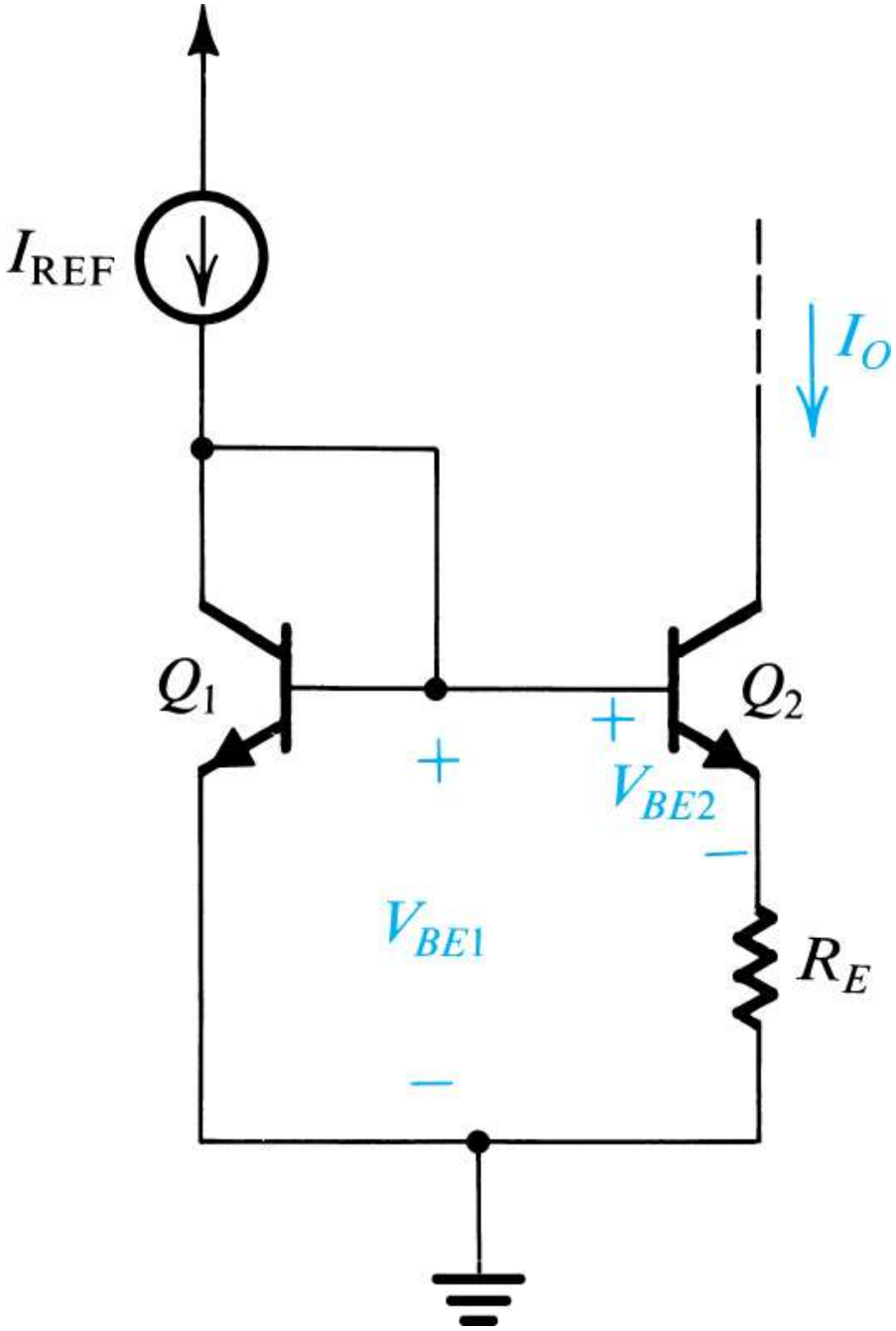


Figure 7.36: The Widlar current source.

7.5. **Current-Mirror Circuits with Improved Performance**

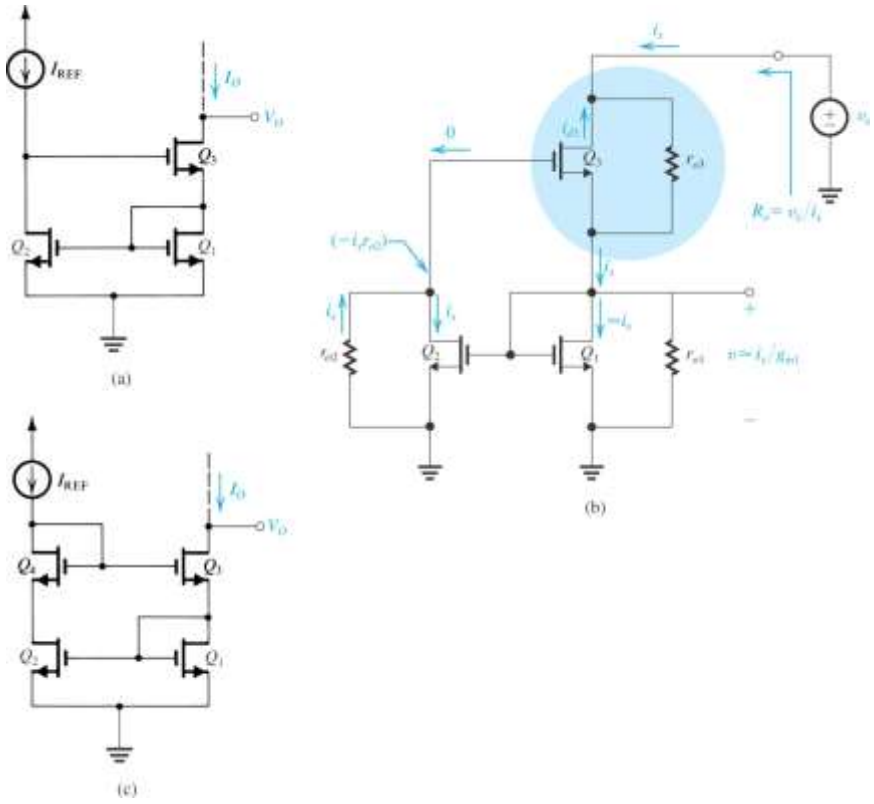


Figure 7.35: The Wilson MOS mirror: **(a)** circuit; **(b)** analysis to determine output resistance; **(c)** modified circuit.

Summary

- Integrated-circuit fabrication technology offers the circuit designer many exciting opportunities, the most important of which is the large number of inexpensive small-area MOS transistors. An overriding concern for IC designers, however, is the minimization of chip area or "silicon real estate." As a result, large-valued resistors and capacitors are virtually absent.
- The basic gain cell of IC amplifier is the CS (CE) amplifier with a current-source load. For an ideal current-source load (i.e. one with infinite output resistance), the transistor operates in an open-circuit fashion and thus provides the maximum gain possible: $A_{vo} = -g_m r_o = -A_0$.

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Summary

- The intrinsic gain A_0 is given by $A_0 = V_A / V_T$ for a BJT and $A_0 = V_A/(V_{OV}/2)$ for a MOSFET. For a BJT, A_0 is constant independent of bias current and device dimensions. For a MOSFET, A_0 is inversely proportional to $I_D^{1/2}$. See equation 7.15.
- Simple current-source loads reduce the gain realized in the basic gain cell because of their finite resistance (usually comparable to the value of r_o of the amplifying transistor).
- To raise the output resistance of the CS or CE transistor, we stack a CG or CB transistor on top. This is cascoding. The CG or CB transistor in the cascode passes the current $g_{m1}v_i$ provided by the CS or CE transistor.

Summary

- A MOS cascode amplifier operating with an ideal current source load achieves a gain of $(g_m r_o)^2 = A_0^2$.
- To realize the full advantage of cascoding, the load current-source must also be cascoded, in which case a gain as high as $1/2A_0^2$ can be obtained.
- Double cascoding is possible in the MOS case only. However, the large number of transistors in the stack between the power-supply rails results in the disadvantages of a severely limited output-signal swing. The folded-cascode configuration helps to resolve this issue.

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Summary

- A CS amplifier with a resistance Rs in its source lead has an output resistance $R_o = (1+g_m R_S)r_o$. The corresponding formula for the BJT case is $R_o = [1+g_m(R_e \| r_\pi)]r_o$.
- Biasing in integrated circuits utilizes current sources. As well, current sources are used as load devices. Typically an accurate and stable reference current is generated and then replicated to provide bias current for the various amplifier stages on the chip. The heart of the current-steering circuitry utilized to perform this function is the current mirror.
- The MOS current mirror has a current transfer ratio of $(W/L)_2/(W/L)_1$. For a bipolar mirror, the ratio is I_{S2}/I_{S1}.

Summary

- Bipolar mirrors suffer from the finite b, which reduces the accuracy of the current transfer ratio.
- Both bipolar and MOS mirrors of the basic type have a finite output resistance equal to r_o of the output device. Also, for proper operation, a voltage of at least 0.3V is required across the output transistor of a simple bipolar mirror ($|V_{OV}|$ for the MOS case).
- Cascoding can be applied to current mirrors to increase their output resistances. An alternative that also solves the β problem is the bipolar case is the Wilson circuit.